

NONSOFC GROUPS EXIST

ANONYMOUS

ABSTRACT. We prove that nonsoc groups exist. The group $\text{EL}_4(L_{\mathbb{F}_2}(1, 2))$ of elementary matrices over the binary Leavitt algebra is infinite, finitely generated, has Kazhdan’s property (T), and is not sofc; the proof is engineered for verification: the two auxiliary matrices on which the construction turns are written as explicit products of elementary matrices, all remaining memberships reduce to one Whitehead identity and the perfectness of Thompson’s group V , and the headline result is independent of K -theory.

The analytic engine is a compression–centralizer criterion. If a Kazhdan group is generated by an infinite Kazhdan subgroup Γ together with finitely many elements that conjugate Γ into itself, then in a hypothetical sofc approximation the expander component decompositions supplied by Kun’s theorem for Γ and for the ambient group can be matched almost bijectively. The matching is conservative: a median normalization of component sizes is almost monotone along compressor arcs, exact conservation under permutations makes the estimate two-sided, and an elementary ℓ^1 co-area inequality on the ambient expanders pins the normalized sizes at their median. Localizing to a single matched component yields a sofc approximation of $K \times J \twoheadrightarrow K$ the compressed image of Γ , J a commuting subgroup—whose K -edges form an essential expander sequence, and the Kun–Thom centralizer obstruction forces J to be LEF. The Leavitt relations realize the scheme with J a copy of Thompson’s group V , which is infinite, simple, and finitely presented, hence not LEF.

Matrix self-similarity of $L = L_k(1, 2)$ spreads the conclusion. For every finite field k , the unit group $L^\times$, every $\text{GL}_r(L)$ with $r \geq 1$, and every $\text{EL}_r(L)$ with $r \geq 2$ are mutually isomorphic infinite finitely generated property (T) nonsoc groups, and the construction can already be carried out in rank two. Finally, a finite multiplication-table construction shows that every finitely generated nonsoc group is a quotient of an infinite finitely presented nonsoc group, Kazhdan whenever the original group is. In particular, not every finitely presented discrete group is sofc.

CONTENTS

1. Introduction	3
Main results	3
Strategy	4

Date: August 2026.

2020 Mathematics Subject Classification. 20F65, 20E26, 20F05, 05C48, 16S88, 19B99.

Key words and phrases. sofc groups, property (T), expander graphs, Leavitt algebras, Thompson’s group, elementary groups.

External inputs, and how to verify this paper	6
Organization	7
2. Sofic approximations, expanders, and LEF groups	7
2.1. Sofic approximations	7
2.2. Generator graphs and edit distance	8
2.3. Expansion and an ℓ^1 co-area estimate	9
2.4. LEF groups	10
2.5. The theorems of Kun and Kun–Thom	10
3. Conservative component matching: the toolkit	12
3.1. Expander refinement	12
3.2. Permutation conservation	13
3.3. Median pinning	13
3.4. Weighted selection	13
3.5. Localization by completion	14
4. The compression–centralizer criterion	15
4.1. Statement	15
4.2. The two component partitions	16
4.3. One-sided refinement along a compressor	17
4.4. Median normalization and conservation	18
4.5. Almost-bijective matching	19
4.6. Selection, localization, and the proof of Theorem 4.1	21
4.7. Two remarks on the mechanism	23
5. Leavitt algebras, elementary groups, and Thompson’s group V	23
5.1. Leaf sets and matrix self-similarity	24
5.2. Elementary groups over rings	25
5.3. Characteristic-two block factorizations	27
5.4. Thompson’s group V in a Leavitt corner	28
6. The compression scheme in adjacent ranks	29
6.1. The comb compressor	30
6.2. The involution z and generation	31
6.3. The commuting corner copy of V	32
6.4. The Leavitt-corner theorem	33
7. The explicit realization: no K -theory	33
8. Rank two and the full panorama	35
8.1. The two-by-two realization	35
8.2. K_1 vanishes and $GL = EL$	37
8.3. Finite fields: all ranks and the unit group	39
9. Finitely presented nonsofic covers	40
10. Concluding remarks	42
Appendix A. An elementary and effective proof of $K_1(L_k(1, 2)) = 0$	43
A.1. The Whitehead class group	43
A.2. Code moves	45
A.3. The degree filtration and width reduction	46
A.4. Windows die	47
A.5. The pencil and the elimination loop	49

A.6. Consequences and effectivity	51
Appendix B. The Lean formalization	52
What the development is	52
How to read the margin notes	52
The trust surface of the formalization	53
How the library is laid out	54
What is not formalized	54
References	55

1. INTRODUCTION

A countable discrete group is *sofic* if every finite portion of its multiplication table admits arbitrarily accurate and asymptotically faithful models by permutations of finite sets. The notion arose in Gromov’s work on symbolic dynamics [10]; Weiss introduced the term and asked whether every countable group is sofic [21]. The class contains all amenable and all residually finite groups, and it is stable under a long list of standard constructions; see [7, 18]. Whether every group is sofic has remained a central open problem of modern group theory; for a recent account of its status, see [8].

We answer the question in the negative.

Main results. Let k be a field and let

$$(1) \quad L = L_k(1, 2) = k\langle s_0, s_1, t_0, t_1 \mid t_i s_j = \delta_{ij}, s_0 t_0 + s_1 t_1 = 1 \rangle$$

be the binary Leavitt algebra [15]. For a unital ring A and $r \geq 2$, write

$$x_{ij}(a) = I_r + aE_{ij} \in \mathrm{GL}_r(A), \quad a \in A, \quad i \neq j,$$

and let $\mathrm{EL}_r(A) \leq \mathrm{GL}_r(A)$ be the subgroup generated by these elementary matrices.

Theorem A. *Let $R = L_{\mathbb{F}_2}(1, 2)$. For every $m \geq 1$, the group*

$$\mathrm{EL}_{m+1}(R)$$

is infinite, finitely generated, has property (T), and is not sofic. In particular $G = \mathrm{EL}_4(R)$ is not sofic, and not every countable discrete group is sofic.

The proof of Theorem A is engineered for audit. The two auxiliary matrices of the construction are exhibited as explicit products of elementary matrices; every remaining elementary membership follows from one explicit Whitehead identity together with the perfectness of Thompson’s group V ; the ranks below four are reached by an explicit isomorphism between the elementary groups of any two ranks ≥ 2 over R (Proposition 5.8); and no K -theory is used.

Theorem B. *Let k be a finite field and let $L = L_k(1, 2)$.*

- (i) *The unit group $L^\times$ is infinite, finitely generated, has property (T), and is not sofic.*

MainResults
universalLeavittEL4_not_
isSofic
universalLeavitt_profile
ambient_profile

MainResults
binaryLeavitt_finiteField_
profile
binaryLeavittUnits_not_
isSofic
binaryLeavittGL_not_isSofic

- (ii) For every $r \geq 1$, every ordered complete binary leaf set $\mathcal{C} = (\alpha_1, \dots, \alpha_r)$ induces an explicit group isomorphism

$$(2) \quad \Theta_{\mathcal{C}} : \mathrm{GL}_r(L) \xrightarrow{\cong} L^{\times}, \quad (a_{ij}) \mapsto \sum_{i,j=1}^r \alpha_i a_{ij} \alpha_j^*.$$

For $r \geq 2$ one has $\mathrm{GL}_r(L) = \mathrm{EL}_r(L)$. Hence all the groups $\mathrm{GL}_r(L)$, $r \geq 1$, and $\mathrm{EL}_r(L)$, $r \geq 2$, are mutually isomorphic, and each is an infinite finitely generated property (T) nonsolic group.

MainResults

exists_finitelyPresented_
nonsolic_group
exists_infinite_
finitelyPresented_nonsolic_
ambient_cover
KazhdanCover
exists_kazhdan_
finitelyPresented_cover_of_
not_isSolic

Theorem C. Every finitely generated nonsolic group is a quotient of an infinite finitely presented nonsolic group. Every nonsolic group with property (T) is a quotient of an infinite finitely presented nonsolic group with property (T). In particular, there is an infinite finitely presented property (T) group H that is not sofic; H may be chosen to surject onto $L^{\times}$, and hence to have a quotient isomorphic to every group listed in Theorem B.

Strategy. The proof has three independent layers.

The first layer is analytic.

CriterionAssembly

isLEF_of_isSolic
not_isSolic_of_not_isLEF

Theorem D (Kazhdan compression–centralizer criterion). Let G be a finitely generated property-(T) group, and let $\Gamma \leq G$ be an infinite finitely generated property-(T) subgroup. Suppose there is a finite set $Q \subseteq G$ with

$$G = \langle \Gamma, Q \rangle \quad \text{and} \quad q\Gamma q^{-1} \leq \Gamma \quad (q \in Q).$$

Fix $q_0 \in Q$, put $K = q_0\Gamma q_0^{-1}$, and let $J \leq \Gamma$ be finitely generated with

$$[K, J] = 1, \quad K \cap J = \{1\}.$$

If G is sofic, then J is LEF.

The point of departure is the tension between the two theorems that govern sofic approximations of Kazhdan groups. Kun’s theorem [13] proves Bowen’s conjecture: every sofic approximation of an infinite property-(T) group is essentially a vertex-disjoint union of uniformly expanding components. The Kun–Thom theorem [14] requires more: if the restriction of a sofic approximation of $\Lambda \times \Delta$ to a Kazhdan factor Λ is essentially one expander, then Δ is locally embeddable into finite groups (LEF). A disjoint union of equal-sized expanders is not enough, because the commuting factor may permute the components—this is exactly what happens in a product approximation.

Compression closes the gap. A hypothetical sofic approximation of G yields a Kun decomposition for Γ and another for G . Transporting the Γ -components by a compressor q gives one-sided inclusions into other Γ -components. Write $s(y)$ for the size of the Γ -component containing y ; on each ambient G -component, normalize by a vertex-weighted median m of the sizes and set

$$(3) \quad f(y) = \frac{s(y)}{s(y) + m}.$$

The one-sided inclusions make f almost monotone along compressor arcs. But a compressor acts by a permutation, so total increase equals total decrease *exactly*. The resulting small ℓ^1 edge variation, fed into an elementary co-area inequality on the ambient expanders, pins f at its median value $1/2$. Hence transported and target components have asymptotically equal sizes, and one-sided containment becomes almost-bijective matching. A weighted diagonal selection extracts a single matched component on which every multiplication, faithfulness, invariance, conjugacy, and graph-edit error is negligible. Restricting and completing the permutations yields a sofic approximation of $K \times J$ whose K -edges form an essential expander sequence, and the Kun–Thom obstruction forces J to be LEF. Theorem D is proved in Section 4 as a corollary of a local statement, Theorem 4.1, which isolates the exact graph-theoretic hypotheses and records precisely what property (T) is used for. Remarks 4.6 and 4.7 test the criterion against sofic examples and show that each hypothesis earns its place.

The second layer is algebraic, and it is where the Leavitt relations enter. Fix $m \geq 2$ and a nontrivial unital ring A carrying a binary Leavitt family—four elements satisfying the two Leavitt relations; no copy of L is assumed. A left-comb binary partition with $m + 1$ leaves produces a compressor $u \in \mathrm{GL}_{m+1}(A)$ that conjugates the block subgroup $\Gamma \cong \mathrm{EL}_m(A)$ into the corner $a \mapsto s_0at_0$ of itself. A block involution z centralizes the compressed image K , so $v = zu$ is a second compressor with the same image, and $vu^{-1} = z$ generates every elementary matrix involving the last coordinate; whenever u and z lie in $\mathrm{EL}_{m+1}(A)$, it follows that $G = \mathrm{EL}_{m+1}(A) = \langle \Gamma, u, v \rangle$. The complementary corner $a \mapsto p_0 + s_1at_1$ carries a commuting copy of Thompson’s group V , which is infinite, simple, and finitely presented, hence not LEF. The corner copy of V is elementary over every ambient ring: V is perfect, and the Whitehead lemma places diagonal products of commutators of units in EL_2 . The only membership hypotheses of the general theorem therefore concern u and z . Over $\mathbb{F}_2$ they are discharged by explicit four-term elementary factorizations, and an explicit rank equivalence between the elementary groups of any two ranks ≥ 2 —block flattening along the self-similarity of the Leavitt relations—then spreads the conclusion from rank four to every rank, which gives Theorem A with no K -theory. Over every finite field the memberships also follow from $K_1(L) = 0$ and the GE property of purely infinite simple rings; the K_1 input itself admits an elementary, effective proof, written out in full in Appendix A. Matrix self-similarity $M_r(L) \cong L$ then spreads nonsoficity and property (T) to the unit group and to all GL_r , and a separate two-by-two realization shows that the scheme already closes in rank two.

The third layer is a finite multiplication-table construction. Every finitely generated nonsofic group G has a finite table fragment that admits no sufficiently accurate permutation model. Imposing precisely that finite table as relators gives a finitely presented group mapping onto G ; the map proves

infinitude and preserves distinctness of the named elements, while the forbidden finite table proves nonsoficity directly. When G is Kazhdan, Shalom’s finite-presentation theorem first supplies a finitely presented Kazhdan cover, and imposing the same table inside its kernel produces a finitely presented Kazhdan nonsofic cover. The covers are proved nonsofic directly, from the forbidden table itself; quotient closure of soficity, which is unknown, plays no role.

External inputs, and how to verify this paper. The proof is organized so that its trust surface is explicit and so that a skeptical reader can audit it in layers.

A machine-checked audit is available as a second, independent layer, and it is reported in the margins: each numbered result below carries a note naming the modules and declarations of the accompanying Lean development that formalize it, or recording that it is formalized only in part, or that it is not formalized and why. The notes are marginal in the strict sense—no argument in this paper appeals to one, and deleting all of them leaves the text unchanged—so a reader uninterested in formalization can ignore them throughout. Appendix B says what the development proves, what it assumes, and where it is stronger than the text.

Theorems A and D rest on the following external inputs: Kun’s expander decomposition theorem [13], the Kun–Thom centralizer obstruction [14], property (T) for elementary groups over finitely generated rings [9], and classical facts about Thompson’s group V —infinite, simple, finitely presented, and 2-generated [11, 6, 5]. Everything else is self-contained. In the proof of Theorem A, the two auxiliary matrices u and z are factored explicitly into elementary matrices, and every remaining elementary membership follows from the explicit Whitehead identity of Lemma 5.5 together with the perfectness of V . The analytic sections use no spectral theory: the only analytic inequality proved and used in the matching argument is the ℓ^1 co-area estimate of Lemma 2.7, established in a few lines from the layer-cake formula, while the approximation-theoretic depth is confined to the cited theorems of Kun and Kun–Thom.

Theorem B additionally uses $K_1(L) = 0$ and the GE property of purely infinite simple rings [1, 3, 2]. The GE property is proved directly in Section 8.2 from strong division, and Remark 8.5 presents an elementary elimination proof of $K_1(L) = 0$, formalized in full in the accompanying Lean development, which removes the K -theoretic input entirely; [2] is retained as an independent confirmation.

Theorem C is self-contained apart from Shalom’s finitely presented Kazhdan covers [19], used only for the Kazhdan refinement.

A reader who wants to check the headline claim should verify, in order: the five finite lemmas of Section 3, each elementary and a few lines long; the proof of Theorem 4.1 in Section 4, which assembles them and invokes Kun’s theorem twice and the Kun–Thom theorem once, in exactly the

forms stated in Section 2.5; the matrix identities of Sections 5 and 6, each a finite computation from the two Leavitt relations; and the two explicit factorizations of Section 7. That path proves Theorem A in full. Sections 8 and 9 lie downstream and can be audited separately.

Organization. Section 2 fixes the sofic, expander, and LEF vocabulary and states the theorems of Kun and Kun–Thom in the forms used. Section 3 isolates the five elementary lemmas of the matching mechanism. Section 4 proves the local compression–centralizer theorem and deduces Theorem D. Section 5 develops Leavitt leaf calculus, matrix self-similarity, the Whitehead lemma, the characteristic-two block factorizations, and the corner realization of V . Section 6 presents the compression scheme in adjacent ranks over an arbitrary ring and proves the Leavitt-corner theorem. Section 7 discharges all memberships explicitly over $\mathbb{F}_2$ and proves Theorem A. Section 8 proves the two-by-two theorem, the equality $\text{GL}_r(L) = \text{EL}_r(L)$, and Theorem B. Section 9 proves Theorem C. Section 10 collects concluding remarks.

2. SOFTIC APPROXIMATIONS, EXPANDERS, AND LEF GROUPS

2.1. Sofic approximations. For $\sigma, \tau \in \text{Sym}(Y)$, let

$$d_H(\sigma, \tau) = \frac{1}{|Y|} |\{y \in Y \mid \sigma(y) \neq \tau(y)\}|$$

denote the normalized Hamming distance.

Definition 2.1. Let G be a countable discrete group. A *sofic approximation* of G is a sequence of maps $\sigma_n : G \rightarrow \text{Sym}(Y_n)$, with $|Y_n| \rightarrow \infty$, such that

- (a) $d_H(\sigma_n(g)\sigma_n(h), \sigma_n(gh)) \rightarrow 0$ for all $g, h \in G$;
- (b) $d_H(\sigma_n(g), \text{id}) \rightarrow 0$ for all $g \in G \setminus \{1\}$.

The group G is *sofic* if it admits a sofic approximation.

The maps σ_n are not assumed to be homomorphisms. Property (b) self-improves: for a sofic group one may always pass to a sofic approximation in which each $\sigma_n(g)$, $g \neq 1$, moves *every* point of a set of density $1 - o(1)$, and it is standard that (a) and (b) force $d_H(\sigma_n(1), \text{id}) \rightarrow 0$. We will use the following exact normalization.

Lemma 2.2 (Finite normalization). *Let $\sigma_n : G \rightarrow \text{Sym}(Y_n)$ be a sofic approximation and let $F \subseteq G$ be a finite symmetric set. Then there is a sofic approximation $\sigma'_n : G \rightarrow \text{Sym}(Y_n)$ with $d_H(\sigma_n, \sigma'_n) \rightarrow 0$ pointwise, such that for all n :*

$$\sigma'_n(1) = \text{id}, \quad \sigma'_n(g^{-1}) = \sigma'_n(g)^{-1} \quad (g \in F).$$

Proof. Set $\sigma'_n(1) = \text{id}$, and note that $d_H(\sigma_n(1), \text{id}) \rightarrow 0$: applying (a) with $g = h = 1$ gives $d_H(\sigma_n(1)^2, \sigma_n(1)) \rightarrow 0$, and composing with the permutation $\sigma_n(1)^{-1}$ preserves d_H .

Choose one representative from each pair $\{g, g^{-1}\} \subseteq F \setminus \{1\}$ with $g \neq g^{-1}$; put $\sigma'_n(g) = \sigma_n(g)$ and $\sigma'_n(g^{-1}) = \sigma_n(g)^{-1}$. Then $d_H(\sigma_n(g^{-1}), \sigma_n(g)^{-1}) \rightarrow 0$,

Sofic
SoficApproximation
IsSofic

Sofic
map_one_close
InverseNormalization
inverseError_negligible

because $\sigma_n(g^{-1})\sigma_n(g)$ is d_H -close to $\sigma_n(1)$, hence to id , and d_H is invariant under composition with the permutation $\sigma_n(g)^{-1}$ on the right.

If $g = g^{-1} \in F \setminus \{1\}$ is an involution, let $A_n = \{y \mid \sigma_n(g)^2 y = y\}$. Since $\sigma_n(g)^2$ is d_H -close to $\sigma_n(g^2) = \sigma_n(1)$, hence to id , we have $|Y_n \setminus A_n| = o(|Y_n|)$. The set A_n is $\sigma_n(g)$ -invariant—if $\sigma_n(g)^2 y = y$, then applying $\sigma_n(g)$ gives $\sigma_n(g)^2(\sigma_n(g)y) = \sigma_n(g)y$ —and $\sigma_n(g)$ restricts to an involution of A_n . Define $\sigma'_n(g)$ to be $\sigma_n(g)$ on A_n and the identity on $Y_n \setminus A_n$; this is an involution of Y_n differing from $\sigma_n(g)$ on $o(|Y_n|)$ points.

Outside $F \cup \{1\}$ put $\sigma'_n = \sigma_n$. Each $\sigma'_n(g)$ differs from $\sigma_n(g)$ on $o(|Y_n|)$ points, so properties (a) and (b) persist. $\square$

The next lemma is the bridge, used many times below, between statements about products of letter permutations and statements about the permutation of the product.

Lemma 2.3 (Word approximation). *Let $\sigma_n : G \rightarrow \text{Sym}(Y_n)$ be a sofie approximation, let $g \in G$, and let $g = t_1 t_2 \cdots t_\ell$ be any fixed factorization with $t_1, \dots, t_\ell \in G$. Then*

$$|\{y \in Y_n \mid \sigma_n(t_1)\sigma_n(t_2) \cdots \sigma_n(t_\ell)y \neq \sigma_n(g)y\}| = o(|Y_n|).$$

Proof. Induction on ℓ , the case $\ell = 1$ being trivial. For $\ell \geq 2$ put $h = t_2 \cdots t_\ell$. Composing with the permutation $\sigma_n(t_1)$ on the left does not change disagreement sets, so $\sigma_n(t_1) \cdots \sigma_n(t_\ell)$ agrees with $\sigma_n(t_1)\sigma_n(h)$ outside the $o(|Y_n|)$ points supplied by the inductive hypothesis; and $\sigma_n(t_1)\sigma_n(h)$ agrees with $\sigma_n(t_1 h) = \sigma_n(g)$ outside $o(|Y_n|)$ points by Definition 2.1(a). The union of the two sets has size $o(|Y_n|)$. $\square$

Whenever an element of a subgroup is written below as a word in a generating set and a property is verified letter by letter, Lemma 2.3 is what transfers the conclusion from the product of the letter permutations to the permutation of the element itself; it is cited explicitly at each such point.

2.2. Generator graphs and edit distance. Let $T \subseteq G$ be a finite set and $\sigma : G \rightarrow \text{Sym}(Y)$ a map. The T -generator graph $X(Y, \sigma, T)$ is the multigraph on the vertex set Y with one edge $\{y, \sigma(t)y\}$ for each pair $(y, t) \in Y \times T$ with $\sigma(t)y \neq y$; parallel edges are retained, fixed points contribute no edge, and every vertex has degree at most $2|T|$. An arc of $t \in T$ is an ordered pair $(y, \sigma(t)y)$.

Conventions. All graphs in this paper are finite multigraphs without loops, and all edge counts—degrees, boundaries, edit distances, and sums over $E(X)$ —count edges with multiplicity. For multigraphs X, X' on the same vertex set Y , the edit distance $\text{ed}(X, X')$ is the minimal number of edge insertions and deletions, with multiplicity, transforming X into X' . A sequence of graphs X_n on Y_n is edit-equivalent to a sequence X'_n if $\text{ed}(X_n, X'_n) = o(|Y_n|)$.

Sofie
word_close
BlockWordCrossing
wordDisagreement_negligible
all_wordCrossing_negligible

For a finite multigraph X and $\emptyset \neq U \subseteq V(X)$, write ∂U for the multiset of edges with exactly one endpoint in U , and

$$h(X) = \min \left\{ \frac{|\partial U|}{|U|} \mid \emptyset \neq U \subseteq V(X), |U| \leq \frac{1}{2}|V(X)| \right\}$$

for the Cheeger constant; for a one-vertex graph the minimum runs over the empty set, and we set $h(X) = +\infty$. A sequence of bounded-degree graphs X_n with $|V(X_n)| \rightarrow \infty$ is an *expander sequence* if $\inf_n h(X_n) > 0$.

Remark 2.4 (Robustness of the conventions). The conventions are robust in exactly the two directions in which they interface with the cited external theorems. Loops are immaterial: a loop lies in no boundary ∂U , so inserting or deleting loops changes no Cheeger constant, no component, and no estimate below; arcs with $\sigma(t)y = y$ are accordingly discarded once and for all in the definition above. Multiplicities are controlled by a uniform bound: in a T -generator graph every unordered pair carries at most $2|T|$ parallel edges, one per arc. For multigraphs of multiplicity at most M : forgetting multiplicities preserves components and connectivity; boundary sizes change by a factor between 1 and M , so a componentwise Cheeger bound h in one convention yields at least h/M in the other; and one simple-graph edit lifts to at most M multigraph edits, so $o(|Y_n|)$ edit-distance statements transfer in both directions. We do not assert any comparison for arbitrary multigraphs—two multigraphs can share an underlying simple graph and differ by linearly many parallel copies; the transfer above uses the multiplicity bound, which holds for every graph in this paper. Consequently every statement below, and each of the two external theorems of Section 2.5, may be read in either convention with constants adjusted; we fix the multiplicity-counting convention throughout.

Definition 2.5. Let (X_n) be a sequence of graphs of uniformly bounded degree on vertex sets Y_n with $|Y_n| \rightarrow \infty$. An *expander decomposition* of (X_n) consists of graphs $\tilde{X}_n$ on Y_n , of uniformly bounded degree, with $\text{ed}(X_n, \tilde{X}_n) = o(|Y_n|)$, together with a constant $h > 0$ such that every connected component of every $\tilde{X}_n$ has Cheeger constant at least h .

Criterion
ExpanderDecomposition

Definition 2.6. A sequence of bounded-degree graphs (X_n) is an *essential expander sequence* if it is edit-equivalent to an expander sequence.

formalized in part

Criterion

MatchingCertificate

EssentialExpanderRepair

induced_expands_eventually

repairedActualGraph_expands_

eventually

bounded as the

matching certificate

DirectedData and its repair,

nonnegative_weighted

2.3. Expansion and an ℓ^1 co-area estimate. The single analytic inequality used in this paper is the following co-area estimate.

Lemma 2.7 (ℓ^1 co-area estimate). *Let X be a finite connected graph and $f : V(X) \rightarrow \mathbb{R}$. Let $c \in \mathbb{R}$ be such that*

$$|\{x \mid f(x) > c\}| \leq \frac{1}{2}|V(X)| \quad \text{and} \quad |\{x \mid f(x) < c\}| \leq \frac{1}{2}|V(X)|.$$

Then

$$\sum_{x \in V(X)} |f(x) - c| \leq \frac{1}{h(X)} \sum_{\{x,y\} \in E(X)} |f(x) - f(y)|.$$

Proof. If X has a single vertex, the median hypotheses force $f \equiv c$ and both sides vanish (with $1/h(X) = 0$ by convention); assume $|V(X)| \geq 2$. Replacing f by $f - c$ we may assume $c = 0$. Write $f = f_+ - f_-$ with $f_{\pm} \geq 0$. For $t > 0$, the level set $U_t = \{x \mid f_+(x) > t\}$ satisfies $|U_t| \leq \frac{1}{2}|V(X)|$, so $|\partial U_t| \geq h(X)|U_t|$ whenever $U_t \neq \emptyset$. By the layer-cake formula,

$$\sum_x f_+(x) = \int_0^\infty |U_t| dt \leq \frac{1}{h(X)} \int_0^\infty |\partial U_t| dt \leq \frac{1}{h(X)} \sum_{\{x,y\} \in E(X)} |f_+(x) - f_+(y)|,$$

since an edge $\{x, y\}$ lies in ∂U_t exactly for t in an interval of length $|f_+(x) - f_+(y)|$. The same bound holds for f_- , and $|f_+(x) - f_+(y)| + |f_-(x) - f_-(y)| \leq |f(x) - f(y)|$ pointwise on edges, because f_+ and f_- have disjoint supports. Adding the two estimates gives the claim. $\square$

2.4. LEF groups. A group J is *LEF* (locally embeddable into finite groups) [20] if for every finite subset $F \subseteq J$ there are a finite group P and a map $\varphi : F \rightarrow P$ that is injective and satisfies $\varphi(g)\varphi(h) = \varphi(gh)$ whenever $g, h, gh \in F$. Every LEF group is sofic; finite and residually finite groups are LEF.

Lemma 2.8. *A finitely presented LEF group is residually finite.*

Proof. Let $J = \langle T \mid W \rangle$ with T, W finite and T symmetric, and let $g \in J \setminus \{1\}$; fix a word w_g in T representing g . Let $F \subseteq J$ be the finite set consisting of 1, the elements of T , all partial products $t_1 \cdots t_i$ of every relator $t_1 \cdots t_\ell \in W$, and all partial products of w_g ; in particular $g \in F$. Let $\varphi : F \rightarrow P$ be a local embedding into a finite group. First, $\varphi(1)\varphi(1) = \varphi(1)$ gives $\varphi(1) = 1_P$. Next, the assignment $t \mapsto \varphi(t)$ kills every relator: its partial products $p_i = t_1 \cdots t_i$ lie in F along with the letters, so $\varphi(p_{i+1}) = \varphi(p_i)\varphi(t_{i+1})$ by multiplicativity, and induction gives $\varphi(t_1) \cdots \varphi(t_\ell) = \varphi(p_\ell) = \varphi(1) = 1_P$. Hence $t \mapsto \varphi(t)$ extends to a homomorphism $\psi : J \rightarrow P$. The same induction along the partial products of w_g , all of which lie in F , gives $\psi(g) = \varphi(g)$, and $\varphi(g) \neq \varphi(1) = 1_P$ by injectivity of φ . Thus g survives in a finite quotient, and J is residually finite. $\square$

2.5. The theorems of Kun and Kun–Thom. We use two structure theorems for sofic approximations of Kazhdan groups.

Theorem 2.9 (Kun [13]). *Let Γ be an infinite finitely generated property-(T) group with a finite symmetric generating set S , and let $\sigma_n : \Gamma \rightarrow \text{Sym}(Y_n)$ be a sofic approximation. Then the S -generator graphs $X(Y_n, \sigma_n, S)$ admit an expander decomposition. Moreover, the modified graphs $\tilde{X}_n$ may be chosen of degree at most d , with the Cheeger constants of all components bounded below by a single constant $h > 0$; the constants d and h depend only on (Γ, S) .*

Kun’s theorem proves Bowen’s conjecture: sofic approximations of infinite Kazhdan groups are essentially disjoint unions of expanders. Theorem 2.9 is deliberately a *one-way* statement: the four-way equivalence of [13, Theorem 3] contains a spectral implication that fails as written and is not imported

not formalized
Not needed: the non-LEF
witness is a direct finite
obstruction, not a
residual-finiteness argument.

KunDecomposition
exists_expanderDecomposition
KunFixedDecomposition
expanderDecomposition
Proved internally, in the
full-sequence one-way form
used here; not assumed.

anywhere in this paper; Remark 2.11 records this precisely, together with two local repairs to the published proofs of the theorems of this subsection. The companion result of Kun and Thom is an obstruction for centralizers.

Theorem 2.10 (Kun–Thom [14]). *Let Λ be an infinite finitely generated property-(T) group with finite symmetric generating set S_Λ , and let Δ be a finitely generated group. Suppose $\Lambda \times \Delta$ admits a sofic approximation $\pi_n : \Lambda \times \Delta \rightarrow \text{Sym}(D_n)$ such that the S_Λ -generator graphs $X(D_n, \pi_n|_\Lambda, S_\Lambda)$ form an essential expander sequence. Then Δ is LEF.*

*KunThomTheorem
isLEF_of_
exactProductExpansion
KunThomEssential
isLEF_of_matchingCertificate
Proved internally, with the
matching-core repair; not
assumed.*

Theorem 2.10 states the theorem of [14] in its *essential* form, which is exactly how it is invoked in Section 4. The statement in [14] assumes that the Λ -graphs form a genuine expander sequence; the essential form is nonetheless the published theorem’s, on two independent grounds. First, the published proof opens under the weaker hypothesis that the sequence is essentially expanding and is carried out under that hypothesis throughout. Second, and independently of any reading of the cited proof, the essential form is proved outright in the accompanying formalization (Appendix B); so the interface used here rests on no charitable interpretation. The heuristic content: a genuine expander is essentially connected, so the commuting factor Δ cannot shuffle components and is forced to act almost by automorphisms of finite structures, which yields LEF approximations.

Interface. The proof in [14] is carried out for *regularly labeled* graphs: each generator label has exactly one outgoing directed edge at every vertex. An approximation normalized as in Lemma 2.2 supplies this structure. Every generator acts by a genuine permutation, and an inverse pair of generators acts by an inverse pair of permutations, so the two directed arcs of a pair combine into one undirected labeled edge; identity loops are discarded by the conventions of Section 2.2. Distinct labels rarely collide: for generators g, h with $h \neq g^{\pm 1}$, a collision $\sigma_n(g)y = \sigma_n(h)y$ makes y a preimage under the permutation $\sigma_n(h)$ of a fixed point of $\sigma_n(h)^{-1}\sigma_n(g)$, which agrees with $\sigma_n(h^{-1}g)$ outside a vanishing fraction of points by approximate multiplicativity and has a vanishing fraction of fixed points by asymptotic faithfulness. These properties are verified for the localized approximation at the end of the proof of Theorem 4.1.

Remark 2.11 (Audit of the two cited proofs). Because the two theorems above carry the entire approximation-theoretic weight of this paper, their published proofs were audited line by line in the course of the formalization, with three findings; we record them here both to delimit exactly what is imported and to credit the arguments of [13, 14], which survive intact.

*KunSpectralCounterexample
switched
rayleigh_testVector
not_isBipartite
The near-bipartite
counterexample to the
spectral implication,
formalized.*

First, Theorem 3 of [13] is stated as a four-way equivalence, and its spectral implication (4) $\Rightarrow$ (1) fails as written: it invokes the Cheeger–Alon–Milman bound, which separates the nonconstant Markov spectrum from $+1$ but not from -1 , and near-bipartite uniform expanders (a two-edge switch in each graph of a bipartite expander family, repeated) satisfy (2)–(4) while violating (1). This does not affect the present paper, nor Theorem 1 of

[13]: the decomposition theorem needs only the forward implications, and Theorem 2.9 above is exactly that one-way statement—the only form used here and the form proved in the formalization.

Second, the proof of the forward direction in [13] normalizes a Markov defect (Lemma 10) without first excluding the degenerate case in which that defect vanishes; the repair separates the zero-defect case, where the displacement estimate holds for homogeneous reasons, before dividing.

Third, the centralizer argument of [14] constructs a correspondence between finite models and later treats it as the graph of a permutation, which it need not yet be; the repair measures the row and column fibers of the correspondence, extracts its injective matching core, and completes that core to a genuine permutation, after which the argument proceeds as published.

Both repairs are local to the proofs and are carried out in full in the accompanying formalization (Appendix B), which proves Theorems 2.9 and 2.10 from first principles; a reader auditing this paper by hand should read [13, 14] with the three points above in mind.

3. CONSERVATIVE COMPONENT MATCHING: THE TOOLKIT

This section isolates five elementary lemmas about finite graphs, permutations, and sequences. They are stated once, in the exact quantitative forms used in Section 4; the only genuinely group-theoretic input of the paper appears there.

3.1. Expander refinement.

Lemma 3.1 (Expander refinement). *Let X be a finite graph, every connected component of which has Cheeger constant at least $h > 0$. Let $V(X) = A_1 \sqcup \cdots \sqcup A_r$ be a partition, and let e be the number of edges of X whose endpoints lie in different parts. Then there is an assignment $C \mapsto i(C)$ of a part to each component of X , with $|C \cap A_{i(C)}|$ maximal, such that*

$$\sum_C |C \setminus A_{i(C)}| \leq \frac{4e}{h}.$$

Proof. Every edge of X lies inside a component, so the crossing edges distribute among the components; let e_C be the number inside C , with $\sum_C e_C = e$. Fix a component C , let $a_1 \geq a_2 \geq \cdots \geq a_k$ be the sizes of its nonempty intersections with the parts, and let $A_{i(C)}$ realize a_1 . We claim

$$|C| - a_1 \leq \frac{4e_C}{h},$$

which proves the lemma upon summation. If C meets only one part, then $C \setminus A_{i(C)} = \emptyset$ and the claim is trivial; assume C meets at least two parts. If a subset $W \subseteq C$ is a union of part-intersections with $|W| \leq \frac{1}{2}|C|$, then every edge of $\partial_{X[C]}W$ joins different parts, so $e_C \geq |\partial_{X[C]}W| \geq h|W|$.

If $a_1 \geq \frac{1}{2}|C|$, apply this to $W = C \setminus A_{i(C)}$, a union of part-intersections of size $|C| - a_1 \leq \frac{1}{2}|C|$; this gives $|C| - a_1 \leq e_C/h$.

Refinement
exists_dominant_cell
le_crossing_of_cell
ComponentRefinement
refineComponent

If $a_1 < \frac{1}{2}|C|$, accumulate part-intersections in any order until the running total first reaches at least $\frac{1}{4}|C|$; the resulting union W satisfies $\frac{1}{4}|C| \leq |W| < \frac{1}{4}|C| + a_1 < \frac{3}{4}|C|$, so $\min(|W|, |C \setminus W|) \geq \frac{1}{4}|C|$, and whichever of $W, C \setminus W$ has size at most $\frac{1}{2}|C|$ is again a union of part-intersections. Applying the boundary bound to it gives $e_C \geq \frac{h}{4}|C| \geq \frac{h}{4}(|C| - a_1)$. $\square$

3.2. Permutation conservation.

Lemma 3.2 (Permutation conservation). *Let Y be a finite set, $\pi \in \text{Sym}(Y)$, and $f : Y \rightarrow \mathbb{R}$. Then*

*PermutationConservation
permutation_conservation_
full*

$$\sum_{y \in Y} (f(y) - f(\pi y))_+ = \sum_{y \in Y} (f(\pi y) - f(y))_+ = \frac{1}{2} \sum_{y \in Y} |f(\pi y) - f(y)|.$$

Proof. Both $\sum_y f(y)$ and $\sum_y f(\pi y)$ equal the same number, so $\sum_y (f(y) - f(\pi y)) = 0$; split into positive and negative parts. $\square$

This exact identity is what converts a one-sided (monotonicity) estimate along the arcs of a permutation into a two-sided ℓ^1 estimate.

3.3. Median pinning.

Lemma 3.3 (Median pinning). *Let Z be a finite graph in which every connected component B satisfies $h(Z[B]) \geq h > 0$. Let $f : V(Z) \rightarrow \mathbb{R}$, and suppose that on every component at most half of the vertices satisfy $f > \frac{1}{2}$ and at most half satisfy $f < \frac{1}{2}$. Then*

*Pinning
median_pinning
ComponentPinning
normalized_pinning_global*

$$\sum_{x \in V(Z)} \left| f(x) - \frac{1}{2} \right| \leq \frac{1}{h} \sum_{\{x,y\} \in E(Z)} |f(x) - f(y)|.$$

Proof. Every edge of Z lies within a component. Apply Lemma 2.7 with $c = \frac{1}{2}$ on each component and sum. $\square$

3.4. Weighted selection.

Lemma 3.4 (Weighted selection). *For each n , let $\{D_{n,i}\}_{i \in I_n}$ be disjoint nonempty subsets of a finite set Y_n , $|Y_n| = N_n \rightarrow \infty$, with $\sum_{i \in I_n} |D_{n,i}| \geq (1 - o(1))N_n$. Suppose that for every fixed M ,*

*Selection
exists_selection
diagonalLevel_error*

$$\sum_{i: |D_{n,i}| \leq M} |D_{n,i}| = o(N_n).$$

Let, for each $r \in \mathbb{N}$, nonnegative error weights $E_{n,i}^{(r)}$ be given with

$$\sum_{i \in I_n} E_{n,i}^{(r)} = o(N_n) \quad \text{for every fixed } r.$$

Then there exist indices $i(n) \in I_n$, for all n in a cofinite set, such that $|D_{n,i(n)}| \rightarrow \infty$ and, for every fixed r ,

$$E_{n,i(n)}^{(r)} = o(|D_{n,i(n)}|).$$

Proof. Replacing $E_{n,i}^{(r)}$ by $\max_{r' \leq r} E_{n,i}^{(r')}$ —which only increases the weights, preserves the hypothesis for each fixed r because the maximum of finitely many families, each of total weight $o(N_n)$, has total weight at most their sum, and only strengthens the conclusion—we may and do assume that $E_{n,i}^{(r)}$ is nondecreasing in r .

For $k \in \mathbb{N}$, call an index $i \in I_n$ *k-bad* if $|D_{n,i}| \leq k$ or $E_{n,i}^{(k)} > \frac{1}{k}|D_{n,i}|$, and *k-good* otherwise; let

$$q_k(n) = \frac{1}{N_n} \sum_{i \text{ k-bad}} |D_{n,i}|.$$

For each fixed k ,

$$q_k(n) \leq \frac{1}{N_n} \left(\sum_{i: |D_{n,i}| \leq k} |D_{n,i}| + k \sum_{i \in I_n} E_{n,i}^{(k)} \right) \xrightarrow{n \rightarrow \infty} 0,$$

the second summand bounding the mass of the second kind of badness by Markov's inequality, and both summands being $o(N_n)$ by hypothesis. Moreover the set of *k-bad* indices is nondecreasing in k : $|D_{n,i}| \leq k$ implies $|D_{n,i}| \leq k+1$, and $E_{n,i}^{(k)} > \frac{1}{k}|D_{n,i}|$ implies $E_{n,i}^{(k+1)} \geq E_{n,i}^{(k)} > \frac{1}{k+1}|D_{n,i}|$, using monotonicity in r . Hence $q_k(n)$ is nondecreasing in k for each fixed n .

Now make one simultaneous choice. Since $q_k(n) \rightarrow 0$ as $n \rightarrow \infty$ for each fixed k , there are integers $n_1 < n_2 < \dots$ with $q_k(n) \leq \frac{1}{k}$ for all $n \geq n_k$. For $n \geq n_1$ put $k_n = \max \{k \mid n_k \leq n\}$, a finite maximum since the n_k strictly increase; then $k_n \rightarrow \infty$ and, by the definition of n_{k_n} , $q_{k_n}(n) \leq \frac{1}{k_n} \rightarrow 0$.

For n large, the k_n -bad indices carry mass $q_{k_n}(n)N_n \leq N_n/k_n = o(N_n)$, while all indices together carry mass at least $(1 - o(1))N_n$; hence some k_n -good index $i(n)$ exists. Then $|D_{n,i(n)}| > k_n \rightarrow \infty$, and for every fixed r : as soon as $k_n \geq r$, monotonicity in r gives

$$E_{n,i(n)}^{(r)} \leq E_{n,i(n)}^{(k_n)} \leq \frac{1}{k_n}|D_{n,i(n)}| = o(|D_{n,i(n)}|). \quad \square$$

3.5. Localization by completion.

Lemma 3.5 (Localization by completion). *Let H be a countable group and $\sigma_n : H \rightarrow \text{Sym}(Y_n)$ maps. Suppose $D_n \subseteq Y_n$ are subsets with $|D_n| \rightarrow \infty$ such that, for every fixed $g, h \in H$:*

- (i) $|\{y \in D_n \mid \sigma_n(g)y \notin D_n\}| = o(|D_n|)$;
- (ii) $|\{y \in D_n \mid \sigma_n(g)\sigma_n(h)y \neq \sigma_n(gh)y\}| = o(|D_n|)$;
- (iii) $|\{y \in D_n \mid \sigma_n(g)y = y\}| = o(|D_n|)$ whenever $g \neq 1$.

For $g \in H$ define $\pi_n(g) \in \text{Sym}(D_n)$ by letting $\pi_n(g)$ agree with $\sigma_n(g)$ on $\{y \in D_n \mid \sigma_n(g)y \in D_n, \sigma_n(g)y \neq \sigma_n(g)y' \forall y' \neq y \in D_n\}$ and extending the resulting partial injection of D_n into itself to a permutation of D_n arbitrarily. Then $\pi_n : H \rightarrow \text{Sym}(D_n)$ is a sofic approximation of H . Moreover, if $g, g^{-1} \in H$ satisfy $\sigma_n(g^{-1}) = \sigma_n(g)^{-1}$, the completions may be chosen with $\pi_n(g^{-1}) = \pi_n(g)^{-1}$, and then

$$|\{y \in D_n \mid \pi_n(g)y \neq \sigma_n(g)y\}| = o(|D_n|).$$

Localization
exists_completion
involutiveCompletion
LocalizedApproximation
toSoficApproximation

Proof. Since $\sigma_n(g)$ is injective on Y_n , the described set consists precisely of the points of D_n sent into D_n by $\sigma_n(g)$, and its complement in D_n has size $o(|D_n|)$ by (i); the partial map is an injection, and any extension to a permutation of D_n differs from $\sigma_n(g)$ on $o(|D_n|)$ points of D_n . Then (ii) transfers: for fixed g, h , the points of D_n where $\pi_n(g)\pi_n(h)y \neq \pi_n(gh)y$ are contained in the union of the $o(|D_n|)$ -sized exceptional sets for g, h, gh (points where $\pi \neq \sigma$, including the $\sigma_n(h)$ -preimage of the exceptional set of g) and the set in (ii); similarly (iii) transfers faithfulness. When $\sigma_n(g^{-1}) = \sigma_n(g)^{-1}$ and $g \neq g^{-1}$, the two partial injections attached to g and to g^{-1} are mutually inverse, so a completion of one can be transported to the other. When $g = g^{-1}$, so that $\sigma_n(g)^2 = \text{id}$, the set $A_{n,g} = \{y \in D_n \mid \sigma_n(g)y \in D_n\}$ is $\sigma_n(g)$ -invariant—if y and $\sigma_n(g)y$ both lie in D_n , then so do $\sigma_n(g)y$ and $\sigma_n(g)^2y = y$ —so $\sigma_n(g)$ restricts to an involution of $A_{n,g}$; extending by the identity on $D_n \setminus A_{n,g}$ yields an involutive completion $\pi_n(g) = \pi_n(g)^{-1}$ that differs from $\sigma_n(g)$ on $o(|D_n|)$ points of D_n by (i). $\square$

4. THE COMPRESSION-CENTRALIZER CRITERION

This section proves the analytic core of the paper. We first state a local version that isolates the exact graph-theoretic hypotheses; property (T) of the ambient group is a sufficient condition for the second hypothesis, and the local statement records precisely what it is used for.

4.1. Statement.

Theorem 4.1 (Local compression–centralizer theorem). *Let G be a finitely generated group and let $\Gamma \leq G$ be an infinite finitely generated property-(T) subgroup with finite symmetric generating set S . Let $Q \subseteq G$ be a finite set with*

*LocalCriterion
LocalCriterionData
CriterionAssembly
isLEF_of_soficApproximation*

$$(4) \quad G = \langle \Gamma, Q \rangle \quad \text{and} \quad q\Gamma q^{-1} \leq \Gamma \quad (q \in Q).$$

Fix $q_0 \in Q$, put $K = q_0\Gamma q_0^{-1}$, and let $J \leq \Gamma$ be finitely generated with

$$(5) \quad [K, J] = 1, \quad K \cap J = \{1\}.$$

Put $A = S \cup Q \cup Q^{-1}$. Suppose G admits a sofic approximation $\sigma_n : G \rightarrow \text{Sym}(Y_n)$ such that

- (i) *the S -generator graphs $X(Y_n, \sigma_n, S)$ admit an expander decomposition, and*
- (ii) *the A -generator graphs $X(Y_n, \sigma_n, A)$ admit an expander decomposition.*

Then J is LEF.

Corollary 4.2 (Kazhdan compression–centralizer criterion; Theorem D). *In the setting of (4) and (5), suppose in addition that G and Γ have property (T). If G is sofic, then J is LEF. Equivalently: if J is not LEF, then G is not sofic.*

*CriterionAssembly
isLEF_of_isSofic
not_isSofic_of_not_isLEF*

Proof. Suppose G is sofic and fix a sofic approximation σ_n . By (4) the finite symmetric set $A = S \cup Q \cup Q^{-1}$ generates G , and G is infinite, containing the infinite subgroup Γ . The restriction $\sigma_n|_\Gamma$ is a sofic approximation of the infinite Kazhdan group Γ , so Theorem 2.9 yields hypothesis (i); applying Theorem 2.9 to G itself with the generating set A yields hypothesis (ii). Theorem 4.1 concludes. $\square$

Remark 4.3. Only the componentwise ℓ^1 median inequality of Lemma 3.3 is extracted from hypothesis (ii); that hypothesis may accordingly be weakened to any decomposition of the A -generator graphs, up to $o(|Y_n|)$ edge edits, into components satisfying a uniform ℓ^1 Poincaré inequality around medians. This separates the approximation-theoretic input, which property (T) supplies through Kun's theorem, from the compression mechanism, which is combinatorial.

The remainder of the section proves Theorem 4.1. Write $N_n = |Y_n|$. All $o(N_n)$ estimates are uniform over the fixed finite data (S, Q, q_0) and the decomposition constants; the set Q itself is never symmetrized inside the compression steps, since the compression hypothesis $q\Gamma q^{-1} \leq \Gamma$ is assumed only for $q \in Q$.

4.2. The two component partitions. Put $S_K = q_0 S q_0^{-1}$, a finite symmetric generating set of K , and fix a finite symmetric generating set S_J of the finitely generated group J . By Lemma 2.2 we may and do assume the exact identities

$$\sigma_n(1) = \text{id}, \quad \sigma_n(g^{-1}) = \sigma_n(g)^{-1} \quad (g \in A \cup S_K \cup S_J),$$

after a pointwise $o(N_n)$ modification; each relevant generator graph changes by $o(N_n)$ edges, so hypotheses (i) and (ii) persist.

By hypothesis (i), fix an expander decomposition of the S -generator graphs: graphs $\tilde{X}_n$ on Y_n of degree at most d_Γ , with

$$\text{ed}(X(Y_n, \sigma_n, S), \tilde{X}_n) = o(N_n),$$

such that every component of $\tilde{X}_n$ has Cheeger constant at least $h_\Gamma > 0$. Let $\mathcal{P}_n$ denote the partition of Y_n into the components of $\tilde{X}_n$, and write $P_n(y)$ for the block containing y . We refer to the blocks of $\mathcal{P}_n$ as Γ -components.

By hypothesis (ii), fix likewise an expander decomposition $\tilde{Z}_n$ of the A -generator graphs, with all components of Cheeger constant at least $h_G > 0$ and uniformly bounded degree; let $\mathcal{B}_n$ be the partition of Y_n into the components of $\tilde{Z}_n$, with blocks $B_n(y)$, the G -components.

Both partitions are almost invariant in the following sense. For every $s \in S$,

$$(6) \quad |\{y \in Y_n \mid P_n(\sigma_n(s)y) \neq P_n(y)\}| = o(N_n),$$

because apart from the $o(N_n)$ edge edits, every arc $(y, \sigma_n(s)y)$ is an edge of $\tilde{X}_n$ or a loop, and edges of $\tilde{X}_n$ do not cross $\mathcal{P}_n$. For every $a \in A$,

$$(7) \quad |\{y \in Y_n \mid B_n(\sigma_n(a)y) \neq B_n(y)\}| = o(N_n),$$

for the same reason applied to $\tilde{Z}_n$.

Both partitions stay fixed from here on: $\mathcal{P}_n$ and $\mathcal{B}_n$ are the exact component partitions of $\tilde{X}_n$ and $\tilde{Z}_n$, and every block keeps its connectivity and its Cheeger constant. This is deliberate—the final step of the proof selects a block of $\mathcal{P}_n$ and uses precisely that it is a genuine component of $\tilde{X}_n$. The two partitions are related only pointwise, in two ways: through the almost-invariance estimates (6) and (7), and through the median normalization of Section 4.4, which compares a point with its compressor image exactly when the two lie in the same G -component; the $o(N_n)$ points where this fails are carried as explicit exceptional sets through every estimate below. In particular, neither partition is assumed to refine the other.

4.3. One-sided refinement along a compressor. Fix $q \in Q$. The group identity $qs = (qsq^{-1})q$ and approximate multiplicativity give, for each $s \in S$, the *conjugacy defect* bound

$$(8) \quad \left| \left\{ y \in Y_n \mid \sigma_n(q)\sigma_n(s)y \neq \sigma_n(qsq^{-1})\sigma_n(q)y \right\} \right| = o(N_n).$$

Indeed, $\sigma_n(q)\sigma_n(s)$ agrees with $\sigma_n(qs)$ outside $o(N_n)$ points, and $\sigma_n(qs) = \sigma_n((qsq^{-1})q)$ agrees with $\sigma_n(qsq^{-1})\sigma_n(q)$ outside $o(N_n)$ points, both by Definition 2.1(a); here qsq^{-1} belongs to the fixed finite set $qSq^{-1} \subseteq \Gamma$, by the compression hypothesis (4).

Transport the graphs $\tilde{X}_n$ by the permutation $\sigma_n(q)$: the transported graph $\sigma_n(q)\tilde{X}_n$ has vertex set Y_n and one edge $\{\sigma_n(q)x, \sigma_n(q)y\}$ for each edge $\{x, y\}$ of $\tilde{X}_n$, multiplicities preserved; its components are $\{\sigma_n(q)C \mid C \in \mathcal{P}_n\}$, with the same component Cheeger constants $\geq h_\Gamma$ and degree bound d_Γ . By (8) and the edit bounds, apart from $o(N_n)$ edge occurrences the edges of the transported graph are of the form $\{\sigma_n(q)y, \sigma_n(\gamma)\sigma_n(q)y\}$ with $\gamma \in qSq^{-1} \subseteq \Gamma$; for each of the finitely many $\gamma \in qSq^{-1} \subseteq \Gamma$, fix a word in S representing it; by Lemma 2.3, $\sigma_n(\gamma)$ agrees with the product of the letter permutations outside $o(N_n)$ points, and by (6) applied to each letter in turn, that product preserves the block $P_n(\cdot)$ outside $o(N_n)$ points. Hence all but $o(N_n)$ of the transported edges lie inside blocks of $\mathcal{P}_n$.

Apply Lemma 3.1 to the transported graph with the partition $\mathcal{P}_n$. This produces a map

$$\Phi_{q,n} : \mathcal{P}_n \rightarrow \mathcal{P}_n, \quad \Phi_{q,n}(C) = \text{the block meeting } \sigma_n(q)C \text{ in maximal size,}$$

with total leakage

$$(9) \quad \sum_{C \in \mathcal{P}_n} e_{q,n}(C) = o(N_n), \quad \text{where} \quad e_{q,n}(C) := |\sigma_n(q)C \setminus \Phi_{q,n}(C)|.$$

Thus each transported Γ -component is almost contained in a single target Γ -component. The containment is a priori one-sided: a large target could still absorb several small transported components. Ruling this out is the business of the median normalization.

4.4. Median normalization and conservation. For $y \in Y_n$, let $s_n(y) = |P_n(y)|$ be the size of its Γ -component. For a G -component $B \in \mathcal{B}_n$, let $m_n(B) \geq 1$ be a *vertex-weighted median* of s_n on B : a value such that

$$|\{y \in B \mid s_n(y) > m_n(B)\}| \leq \frac{1}{2}|B| \quad \text{and} \quad |\{y \in B \mid s_n(y) < m_n(B)\}| \leq \frac{1}{2}|B|.$$

Define the *median normalization*

$$(10) \quad f_n(y) = \frac{s_n(y)}{s_n(y) + m_n(B_n(y))} \in (0, 1).$$

Since $t \mapsto t/(t+m)$ is strictly increasing, on every G -component B at most half the vertices satisfy $f_n > \frac{1}{2}$ and at most half satisfy $f_n < \frac{1}{2}$; that is, $\frac{1}{2}$ is a median of f_n on every component of $\tilde{Z}_n$.

Variation along S . For $s \in S$,

$$(11) \quad \sum_{y \in Y_n} |f_n(\sigma_n(s)y) - f_n(y)| = o(N_n),$$

since the summand vanishes unless $P_n(\sigma_n(s)y) \neq P_n(y)$ or $B_n(\sigma_n(s)y) \neq B_n(y)$, a set of size $o(N_n)$ by (6) and (7), and $0 < f_n < 1$.

One-sided variation along $q \in Q$. Call $y \in Y_n$ a *good point* for q if

$$(12) \quad \sigma_n(q)y \in \Phi_{q,n}(P_n(y)) \quad \text{and} \quad B_n(\sigma_n(q)y) = B_n(y).$$

Let y be good, $C = P_n(y)$, $D = \Phi_{q,n}(C)$, and $e = e_{q,n}(C)$; then

$$s_n(\sigma_n(q)y) = |D| \geq |\sigma_n(q)C \cap D| = |C| - e = s_n(y) - e.$$

Since both y and $\sigma_n(q)y$ lie in the same G -component B , they are normalized by the same median $m = m_n(B) \geq 1$. For $a \geq (1-\delta)a' \geq 0$ and $m \geq 1$,

$$(13) \quad \frac{a'}{a' + m} - \frac{a}{a + m} \leq \frac{a'}{a' + m} - \frac{(1-\delta)a'}{(1-\delta)a' + m} \leq \delta,$$

where the first inequality is monotonicity of $t \mapsto t/(t+m)$ and the second is a direct computation: the difference equals $\delta a' m / ((a' + m)((1-\delta)a' + m)) \leq \delta$. Applying (13) with $a' = |C|$, $a = |D| \geq |C| - e$, $\delta = e/|C|$ gives, for every good point y ,

$$(f_n(y) - f_n(\sigma_n(q)y))_+ \leq \frac{e_{q,n}(P_n(y))}{|P_n(y)|}.$$

Summing over Y_n —good points of a block C contribute at most $e_{q,n}(C)$ in total, points with $\sigma_n(q)y \notin \Phi_{q,n}(P_n(y))$ number $e_{q,n}(C)$ per block and contribute at most 1 each, and points whose arc crosses $\mathcal{B}_n$ contribute at most 1 each—we obtain

$$(14) \quad \begin{aligned} \sum_{y \in Y_n} (f_n(y) - f_n(\sigma_n(q)y))_+ &\leq 2 \sum_{C \in \mathcal{P}_n} e_{q,n}(C) + |\{y \mid B_n(\sigma_n(q)y) \neq B_n(y)\}| \\ &= o(N_n) \end{aligned}$$

by (9) and (7). The compressed image can only grow—but growth is globally impossible:

Conservation. Since $\sigma_n(q)$ is a permutation of Y_n , Lemma 3.2 upgrades (14) to the two-sided estimate

$$(15) \quad \sum_{y \in Y_n} |f_n(\sigma_n(q)y) - f_n(y)| = 2 \sum_{y \in Y_n} (f_n(y) - f_n(\sigma_n(q)y))_+ = o(N_n) \quad (q \in Q),$$

and the change of variables $y = \sigma_n(q)x$, legitimate because $\sigma_n(q^{-1}) = \sigma_n(q)^{-1}$, extends (15) to all $q \in Q \cup Q^{-1}$.

Pinning. Every edge of the A -generator graph $X(Y_n, \sigma_n, A)$ arises from an arc $(y, \sigma_n(a)y)$ with $a \in A = S \cup Q \cup Q^{-1}$, and each edge arises from at most $2|A|$ arcs. Hence (11) and (15) give

$$\sum_{\{x,y\} \in E(X(Y_n, \sigma_n, A))} |f_n(x) - f_n(y)| = o(N_n).$$

Since $0 < f_n < 1$, editing $o(N_n)$ edges changes this sum by at most $o(N_n)$; therefore the same bound holds over $E(\tilde{Z}_n)$. As $\frac{1}{2}$ is a median of f_n on every component of $\tilde{Z}_n$ and all component Cheeger constants are at least h_G , Lemma 3.3 yields the concentration estimate

$$(16) \quad \sum_{y \in Y_n} \left| f_n(y) - \frac{1}{2} \right| = o(N_n).$$

In words: after median normalization, almost every Γ -component has size asymptotically equal to the median size on its G -component, because $s/(s+m) \rightarrow \frac{1}{2}$ exactly when $s/m \rightarrow 1$.

4.5. Almost-bijective matching.

Proposition 4.4 (Matching). *Fix $q \in Q$. There are subcollections $\mathcal{G}_{q,n} \subseteq \mathcal{P}_n$ such that:*

- (i) $\sum_{C \in \mathcal{P}_n \setminus \mathcal{G}_{q,n}} |C| = o(N_n)$;
- (ii) for every $C \in \mathcal{G}_{q,n}$, with $D = \Phi_{q,n}(C)$,
- (17) $|\sigma_n(q)C \triangle D| = o(1) \cdot |C|$ uniformly over $C \in \mathcal{G}_{q,n}$;
- (iii) for n large, $\Phi_{q,n}$ is injective on $\mathcal{G}_{q,n}$;
- (iv) $\sum_{C \in \mathcal{G}_{q,n}} |\Phi_{q,n}(C)| = (1 - o(1)) N_n$.

*MedianNormalization
medianNormalize
MatchingPreparation
acceptable_symmDiff_le
Criterion
symmDiff_le_of_pinned
matching_injective*

Proof. Choose $\eta_n \downarrow 0$ so slowly that the three global errors—in (7) for the element q , in (9), and in (16)—are all $o(\eta_n^2 N_n)$. Let $W_n \subseteq Y_n$ be the union of the four sets

$$\begin{aligned} & \left\{ y \mid \left| f_n(y) - \frac{1}{2} \right| > \eta_n \right\}, \quad \sigma_n(q)^{-1} \left\{ y \mid \left| f_n(y) - \frac{1}{2} \right| > \eta_n \right\}, \\ & \{ y \mid B_n(\sigma_n(q)y) \neq B_n(y) \}, \quad \{ y \mid \sigma_n(q)y \notin \Phi_{q,n}(P_n(y)) \}. \end{aligned}$$

Markov's inequality applied to (16) bounds the first set by $\eta_n^{-1} \cdot o(\eta_n^2 N_n) = o(\eta_n N_n)$; the second is a bijective preimage of the first; the last two have sizes $o(\eta_n^2 N_n)$ by the choice of η_n . Hence $|W_n| = o(\eta_n N_n)$.

Discard from $\mathcal{P}_n$ every block C with $e_{q,n}(C) > \eta_n|C|$ or $|C \cap W_n| > \eta_n|C|$, and call the remaining collection $\mathcal{G}_{q,n}$. The discarded mass is at most $\eta_n^{-1} \sum_C e_{q,n}(C) + \eta_n^{-1}|W_n| = o(N_n)$, proving (i).

Let $C \in \mathcal{G}_{q,n}$ and pick $y \in C \setminus W_n$, possible since $\eta_n < 1$. Put $D = \Phi_{q,n}(C)$ and $B = B_n(y) = B_n(\sigma_n(q)y)$, and write $m = m_n(B)$. Since $y \notin W_n$, we have $\sigma_n(q)y \in D$, so $s_n(\sigma_n(q)y) = |D|$, and

$$\left| \frac{|C|}{|C| + m} - \frac{1}{2} \right| \leq \eta_n, \quad \left| \frac{|D|}{|D| + m} - \frac{1}{2} \right| \leq \eta_n.$$

Solving $\left| \frac{a}{a+m} - \frac{1}{2} \right| \leq \eta$ gives $\frac{1-2\eta}{1+2\eta} m \leq a \leq \frac{1+2\eta}{1-2\eta} m$; applying this to $a = |C|$ and to $a = |D|$ and dividing,

$$(18) \quad |D| = (1 + O(\eta_n)) |C| \quad \text{uniformly over } C \in \mathcal{G}_{q,n}.$$

With $e = e_{q,n}(C) \leq \eta_n|C|$,

$$\begin{aligned} |\sigma_n(q)C \triangle D| &= |\sigma_n(q)C \setminus D| + |D \setminus \sigma_n(q)C| \\ &\leq e + (|D| - |C| + e) \leq 2e + ||D| - |C|| = O(\eta_n)|C|, \end{aligned}$$

proving (ii).

For (iii): if $C \neq C'$ in $\mathcal{G}_{q,n}$ had $\Phi_{q,n}(C) = \Phi_{q,n}(C') = D$, the disjoint sets $\sigma_n(q)C \cap D$ and $\sigma_n(q)C' \cap D$ would, by (17) and (18), each occupy a fraction $1 - o(1)$ of D , uniformly—impossible for n large.

For (iv): by (i), (ii), and (18),

$$\sum_{C \in \mathcal{G}_{q,n}} |\Phi_{q,n}(C)| = (1+o(1)) \sum_{C \in \mathcal{G}_{q,n}} |C| = (1+o(1))(1-o(1))N_n = (1-o(1))N_n.$$

□

Thus one-sided refinement, conservation, and pinning together force the compressor to permute the Γ -components almost bijectively, with asymptotically equal sizes. One more observation is needed before selecting a single matched pair.

Lemma 4.5. *For every fixed M , $\sum \{|C| : C \in \mathcal{P}_n, |C| \leq M\} = o(N_n)$.*

Proof. Let $g_1, \dots, g_{M+1} \in \Gamma$ be distinct elements; they exist since Γ is infinite. For all but $o(N_n)$ points y the points $\sigma_n(g_1)y, \dots, \sigma_n(g_{M+1})y$ are pairwise distinct: for $i \neq j$, $\sigma_n(g_i)$ agrees with $\sigma_n(g_i g_j^{-1})\sigma_n(g_j)$ outside $o(N_n)$ points by Definition 2.1(a), and $\sigma_n(g_i g_j^{-1})$ moves $1 - o(1)$ of all points by Definition 2.1(b). Moreover, fixing a word in S representing each g_i , Lemma 2.3 together with (6) applied along its letters shows that for all but $o(N_n)$ points y all these points lie in $P_n(y)$. A point y satisfying both conditions has $|P_n(y)| \geq M + 1$; hence the mass of blocks of size at most M is $o(N_n)$. □

ComponentDivergence
smallBlockVertices_
negligible

4.6. Selection, localization, and the proof of Theorem 4.1.

Proof of Theorem 4.1. Apply Proposition 4.4 with $q = q_0$. For n large this gives a collection of *matched pairs*

$$(19) \quad \{(C, D) : C \in \mathcal{G}_{q_0, n}, D = \Phi_{q_0, n}(C)\},$$

with pairwise distinct sources, pairwise distinct targets, target mass $(1 - o(1))N_n$, and $|\sigma_n(q_0)C \triangle D| = o(1)|C|$ uniformly.

Fix an exhaustion $L_1 \subseteq L_2 \subseteq \dots$ of the countable group $KJ \leq \Gamma$ by finite subsets with $\bigcup_r L_r = KJ$, each L_r containing 1, the finite sets S_K and S_J , and closed under inverses, and such that $L_r \cdot L_r \subseteq L_{r+1}$. To each matched pair (C, D) and each r attach the error weight $E^{(r)}(C, D)$, the sum of:

- (a) $|\sigma_n(q_0)C \triangle D|$;
- (b) the number of edges of $\tilde{X}_n$ with an endpoint in C or D that are edited relative to the S -generator graph, plus the number of edges of the transported graph $\sigma_n(q_0)\tilde{X}_n$ with an endpoint in $D \cup \sigma_n(q_0)C$ that are edited relative to the transported S -generator graph;
- (c) $|\{y \in C \mid \sigma_n(q_0)\sigma_n(s)y \neq \sigma_n(q_0sq_0^{-1})\sigma_n(q_0)y \text{ for some } s \in S\}|$;
- (d) $|\{y \in D \mid \sigma_n(g)\sigma_n(h)y \neq \sigma_n(gh)y \text{ for some } g, h \in L_r\}|$;
- (e) $|\{y \in D \mid \sigma_n(g)y = y \text{ for some } g \in L_r \setminus \{1\}\}|$;
- (f) $|\{y \in D \mid P_n(\sigma_n(g)y) \neq P_n(y) \text{ for some } g \in L_r\}|$.

For every fixed r , the sum of $E^{(r)}$ over the matched pairs is $o(N_n)$: (a) by Proposition 4.4(ii) and (i); (b): the sources are pairwise disjoint, the targets are pairwise disjoint, and a fixed block can occur at most once as a source and at most once as a target; hence each endpoint of an edge lies in at most one source and at most one target, and, an edge having two endpoints, each edited edge is counted for at most four matched pairs—likewise in the transported graph, where the sets $\sigma_n(q_0)C$ are pairwise disjoint images of the disjoint sources—while the global edit counts are $o(N_n)$; (c) by (8) over the finite set S , the sources being disjoint; (d) and (e) by soficity of σ_n over the finite set $L_r \cup L_r^2$, the targets being disjoint; (f) by Lemma 2.3 together with (6) applied along the letters of a fixed word in S representing each $g \in L_r \subseteq \Gamma$, the targets being disjoint.

The targets D are pairwise disjoint blocks of $\mathcal{P}_n$ with total mass $(1 - o(1))N_n$, and by Lemma 4.5 the total mass of targets of size at most M is $o(N_n)$ for every fixed M . Lemma 3.4 therefore produces matched pairs (C_n, D_n) with

$$(20) \quad |D_n| \rightarrow \infty, \quad E^{(r)}(C_n, D_n) = o(|D_n|) \quad \text{for every fixed } r.$$

Put $U_n = \sigma_n(q_0)C_n$, so $|U_n \triangle D_n| = o(|D_n|)$ and $|U_n| = |C_n| = (1 + o(1))|D_n|$. Define

$$(21) \quad D'_n = U_n,$$

a set of size $|C_n| \rightarrow \infty$ that agrees with D_n up to $o(|D_n|)$ points. Every error weight in (20) transfers from the pair (C_n, D_n) to D'_n at a cost of

$o(|D'_n|)$, by separate accounting: the point-counted errors (d), (e), (f) are counted over D_n , and replacing D_n by D'_n changes each count by at most $|D_n \triangle D'_n| = o(|D'_n|)$; the error (c) is counted over C_n , which is unchanged; the edge-counted errors (b) concern graphs of degree at most $\max(d_\Gamma, 2|S|)$, so replacing D_n by D'_n in the incidence condition changes those counts by at most that degree bound times $|D_n \triangle D'_n| = o(|D'_n|)$; and the quantity (a) does not change.

Localization. Consider the countable group $H = KJ \leq \Gamma$. By (5), K and J commute and intersect trivially, so

$$H = KJ \cong K \times J.$$

For every fixed $g \in H$, the transferred error (f) shows that $\sigma_n(g)$ maps all but $o(|D'_n|)$ points of D'_n into D_n . Injectivity upgrades D_n to D'_n : since $\sigma_n(g)$ is a permutation of Y_n , at most $|D_n \setminus D'_n| = o(|D'_n|)$ points of D'_n can land in $D_n \setminus D'_n$, so $\sigma_n(g)$ maps all but $o(|D'_n|)$ points of D'_n into D'_n itself. The transferred errors (d) and (e) give multiplicativity and faithfulness on D'_n for every fixed pair $g, h \in H$, by choosing r with $g, h \in L_r$. Lemma 3.5, with completions on the symmetric set $S_K \cup S_J$ chosen compatibly with the exact inverse identities of Section 4.2, yields a sofic approximation

$$\pi_n : K \times J \longrightarrow \text{Sym}(D'_n)$$

with $|\{y \in D'_n \mid \pi_n(g)y \neq \sigma_n(g)y\}| = o(|D'_n|)$ for every fixed g , and $\pi_n(g^{-1}) = \pi_n(g)^{-1}$ for $g \in S_K \cup S_J$.

The K -edges expand. Restrict $\tilde{X}_n$ to C_n and transport by $\sigma_n(q_0)$: the resulting graph Ξ_n on $D'_n = \sigma_n(q_0)C_n$ is connected with Cheeger constant at least h_Γ and degree at most d_Γ , because C_n is a component of $\tilde{X}_n$. We claim that Ξ_n and the S_K -generator graph $X(D'_n, \pi_n, S_K)$ differ by $o(|D'_n|)$ edges. Indeed, an edge of either graph that fails to be an edge of the other must involve one of the following, each affecting $o(|D'_n|)$ edge occurrences by (20):

- (1) an edge of $\tilde{X}_n[C_n]$ edited relative to the S -generator graph on C_n , or conversely (error (b));
- (2) an arc $(y, \sigma_n(s)y)$, $y \in C_n$, whose $\sigma_n(q_0)$ -transport disagrees with the $\sigma_n(q_0 s q_0^{-1})$ -arc at $\sigma_n(q_0)y$ (error (c));
- (3) a point of D'_n where $\pi_n(g) \neq \sigma_n(g)$ for some $g \in S_K$ (localization defect);
- (4) an arc of some $g \in S_K$ leaving D'_n (transferred error (f) and the symmetric difference (a)).

Hence the S_K -generator graphs of $\pi_n|_K$ form an essential expander sequence in the sense of Definition 2.6.

Conclusion. The group $K = q_0 \Gamma q_0^{-1} \cong \Gamma$ is an infinite finitely generated property-(T) group with symmetric generating set S_K , the group J is finitely generated, and π_n is a sofic approximation of $K \times J$ whose S_K -generator graphs form an essential expander sequence. The regular-labeling requirements recorded after Theorem 2.10 hold for π_n as well: each $\pi_n(g)$ is a genuine permutation of D'_n , with $\pi_n(g^{-1}) = \pi_n(g)^{-1}$ for $g \in S_K \cup S_J$; and

for $g, h \in S_K \cup S_J$ with $h \neq g^{\pm 1}$, the collision set $\{y \in D'_n \mid \pi_n(g)y = \pi_n(h)y\}$ has size $o(|D'_n|)$, by the transferred errors (d) and (e) applied to the pair (h^{-1}, g) and the element $h^{-1}g \in L_2 \setminus \{1\}$, together with the $o(|D'_n|)$ agreement between π_n and σ_n on $S_K \cup S_J \cup \{h^{-1}g\}$. Theorem 2.10 applies and shows that J is LEF. $\square$

4.7. Two remarks on the mechanism.

Remark 4.6 (The product obstruction). The compression hypothesis cannot be deleted from Theorem D. For a direct product $\Lambda \times \Delta$ of an infinite Kazhdan group with a residually finite Kazhdan group, product sofic approximations exist whenever Λ is sofic, and in them the Δ -factor genuinely permutes the Λ -components: the Λ -component partition has all blocks of equal size, and no single component is almost invariant under the second factor. The role of the compressors $q \in Q$, through (4), is to generate the whole ambient group while conjugating Γ into itself, so that the ambient expansion of hypothesis (ii) couples the component structure of Γ to itself across all of G and forbids exactly this shuffling.

Remark 4.7 (The hypothesis $J \leq \Gamma$ is essential). The subgroup J is required to lie in Γ , and this is used twice: through (6) in error (f), which keeps the J -arcs inside the matched component, and through the localization step, which restricts σ_n on elements of $KJ \leq \Gamma$. The hypothesis cannot be dropped in favor of $J \leq G$. Indeed, groups satisfying all the remaining hypotheses with a non-LEF subgroup $J \leq G$ commuting with K exist in abundance among sofic groups—already $J = \text{BS}(2, 3) = \langle a, t \mid ta^2t^{-1} = a^3 \rangle$ is a finitely presented group that is known to be sofic and is non-Hopfian [4], therefore not residually finite, therefore not LEF by Lemma 2.8. The criterion is calibrated exactly so that the commuting subgroup rides inside the compressed expander, and this is what the Leavitt corner construction of Sections 6–7 arranges algebraically.

5. LEAVITT ALGEBRAS, ELEMENTARY GROUPS, AND THOMPSON’S GROUP V

Throughout, k is a field and $L = L_k(1, 2)$ is the binary Leavitt algebra (1), with generators s_0, s_1, t_0, t_1 and relations $t_i s_j = \delta_{ij}$, $s_0 t_0 + s_1 t_1 = 1$. Everything in this section is a consequence of the two relations alone, and it costs nothing to say so once and for all.

Definition 5.1. A *binary Leavitt family* in a unital ring A is a quadruple (s_0, s_1, t_0, t_1) of elements of A satisfying

$$t_i s_j = \delta_{ij} \quad (i, j \in \{0, 1\}) \quad \text{and} \quad s_0 t_0 + s_1 t_1 = 1.$$

The generators of L form a binary Leavitt family in L , and a unital homomorphic image of L carries one; but a family is strictly less than a copy of L —no injectivity is assumed—and the statements below that mention

Leavitt
LeavittFamily

only a family are correspondingly stronger than their specializations to L . Fix a unital ring A carrying a binary Leavitt family, and write

$$p_0 = s_0 t_0, \quad p_1 = s_1 t_1,$$

a pair of orthogonal idempotents summing to 1: $p_i p_j = \delta_{ij} p_i$, since $p_0 p_1 = s_0(t_0 s_1)t_1 = 0$ and $p_i^2 = s_i(t_i s_i)t_i = p_i$. For a binary word $w = w_1 \cdots w_\ell \in \{0, 1\}^*$, put $s_w = s_{w_1} \cdots s_{w_\ell}$ and $s_w^* = t_{w_\ell} \cdots t_{w_1}$, so that $s_w^* s_{w'} = \delta_{w, w'}$ whenever $|w| = |w'|$, and more generally $s_w^* s_{w'}$ vanishes unless one of w, w' is a prefix of the other.

Two remarks on size. The algebra L itself is infinite-dimensional over k : the elements $s_0^i s_1$, $i \geq 0$, are killed on the left by mutually orthogonal projections and are linearly independent. And a family forces infinitude with no reference to L :

Lemma 5.2. *A nontrivial unital ring A carrying a binary Leavitt family is infinite.*

Proof. The idempotents $e_i = s_0^i t_0^i$, $i \geq 0$, are pairwise distinct. Indeed, suppose $e_i = e_j$ with $i < j$. Multiplying on the left by t_0^i and on the right by s_0^i , and using $t_0^i s_0^i = 1$, gives $1 = s_0^d t_0^d$ with $d = j - i \geq 1$. Then $t_1 = t_1 \cdot 1 = (t_1 s_0) s_0^{d-1} t_0^d = 0$, whence $1 = t_1 s_1 = 0$, contradicting nontriviality. $\square$

5.1. Leaf sets and matrix self-similarity. A finite set of binary words $\mathcal{C} = (\alpha_1, \dots, \alpha_r)$ (we use the same symbols for the words and for the corresponding elements $s_{\alpha_i} \in A$) is a *complete leaf set* if the words are the leaves of a finite full rooted binary tree; equivalently, if they are pairwise prefix-incomparable and the cylinders they define partition the boundary $\{0, 1\}^{\mathbb{N}}$.

Lemma 5.3 (Leaf identities). *Let $\mathcal{C} = (\alpha_1, \dots, \alpha_r)$ be a complete leaf set. Then in every unital ring A carrying a binary Leavitt family:*

$$\alpha_i^* \alpha_j = \delta_{ij} \quad \text{and} \quad \sum_{i=1}^r \alpha_i \alpha_i^* = 1.$$

Proof. Prefix-incomparability and the relations $t_i s_j = \delta_{ij}$ give the first identity. The second holds by induction on the tree: for the trivial tree it reads $1 = 1$; replacing a leaf α by its two children $\alpha 0, \alpha 1$ changes the sum by

$$s_{\alpha 0} s_{\alpha 0}^* + s_{\alpha 1} s_{\alpha 1}^* - s_{\alpha} s_{\alpha}^* = s_{\alpha} (s_0 t_0 + s_1 t_1 - 1) s_{\alpha}^* = 0. \quad \square$$

Proposition 5.4 (Matrix self-similarity). *Let A be a unital ring carrying a binary Leavitt family and let $\mathcal{C} = (\alpha_1, \dots, \alpha_r)$ be an ordered complete leaf set. Then*

$$\Theta_{\mathcal{C}} : M_r(A) \longrightarrow A, \quad (a_{ij}) \longmapsto \sum_{i,j=1}^r \alpha_i a_{ij} \alpha_j^*,$$

is an isomorphism of unital rings—of unital k -algebras when A is a k -algebra—with inverse $x \mapsto (\alpha_i^ x \alpha_j)_{ij}$. It restricts to group isomorphisms $\mathrm{GL}_r(A) \cong A^\times$ for all $r \geq 1$. In particular $M_r(L) \cong L$ for every $r \geq 1$.*

Leavitt
infinite
s0_pow_injective

LeavittWords
wordT_mul_wordS_self
wordT_mul_wordS_of_
incomparable
cylinder_split

MatrixSelfSimilarity
ringEquivMatrix
LeavittSelfSimilarity
binaryMatrixRingEquiv
Every positive rank, with an
explicit inverse.

Proof. The map Θ_C is additive—and k -linear when A is a k -algebra—sends the identity matrix to $\sum_i \alpha_i \alpha_i^* = 1$ by Lemma 5.3, and is multiplicative:

$$\Theta_C(a) \Theta_C(b) = \sum_{i,j,l,m} \alpha_i a_{ij} \alpha_j^* \alpha_l b_{lm} \alpha_m^* = \sum_{i,m} \alpha_i \left(\sum_j a_{ij} b_{jm} \right) \alpha_m^* = \Theta_C(ab),$$

using $\alpha_j^* \alpha_l = \delta_{jl}$. The displayed inverse is two-sided by the same identities: $\alpha_i^* \Theta_C(a) \alpha_j = a_{ij}$, and $\sum_{i,j} \alpha_i (\alpha_i^* x \alpha_j) \alpha_j^* = 1 \cdot x \cdot 1 = x$. An algebra isomorphism restricts to an isomorphism of unit groups, and $\mathrm{GL}_r(L) = M_r(L)^\times$. $\square$

For every $r \geq 2$ there is a canonical choice, the *left comb*

$$(22) \quad C_r = (s_1, s_0 s_1, s_0^2 s_1, \dots, s_0^{r-2} s_1, s_0^{r-1}),$$

the leaf set of the tree in which every left child is subdivided; the ordering lists the leaves by depth with the leftmost branch last.

5.2. Elementary groups over rings. Let A be a unital ring. Recall $x_{ij}(a) = I_r + aE_{ij}$ for $i \neq j$, the relations $x_{ij}(a)x_{ij}(b) = x_{ij}(a+b)$, and the Steinberg commutator relation

$$[x_{il}(a), x_{lj}(b)] = x_{ij}(ab) \quad (i, j, l \text{ distinct}),$$

where $[g, h] = ghg^{-1}h^{-1}$.

Lemma 5.5 (Whitehead commutator lemma). *Let A be a unital ring and $a, b \in A^\times$. Put*

$$w(a) = x_{12}(a) x_{21}(-a^{-1}) x_{12}(a) = \begin{pmatrix} 0 & a \\ -a^{-1} & 0 \end{pmatrix}.$$

Then

$$\mathrm{diag}(a, a^{-1}) = w(a) w(-1) \in \mathrm{EL}_2(A),$$

and

$$\mathrm{diag}(aba^{-1}b^{-1}, 1) = \mathrm{diag}(a, a^{-1}) \mathrm{diag}(b, b^{-1}) \mathrm{diag}((ba)^{-1}, ba) \in \mathrm{EL}_2(A).$$

Consequently, for every $m \geq 2$, every product of commutators of units $c \in [A^\times, A^\times]$, and every diagonal position, the matrix with c at that position and 1 elsewhere on the diagonal lies in $\mathrm{EL}_m(A)$.

Proof. First, $x_{12}(a)x_{21}(-a^{-1}) = \begin{pmatrix} 1-aa^{-1} & a \\ -a^{-1} & 1 \end{pmatrix} = \begin{pmatrix} 0 & a \\ -a^{-1} & 1 \end{pmatrix}$, and multiplying by $x_{12}(a)$ on the right adds the first column, multiplied on the right by a , to the second, giving $w(a) = \begin{pmatrix} 0 & a \\ -a^{-1} & 0 \end{pmatrix}$. Then

$$w(a)w(-1) = \begin{pmatrix} 0 & a \\ -a^{-1} & 0 \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} a & 0 \\ 0 & a^{-1} \end{pmatrix}.$$

The displayed product of three diagonal matrices has first entry $ab(ba)^{-1} = aba^{-1}b^{-1}$ and second entry $a^{-1}b^{-1}(ba) = 1$. For the last statement, embed $\mathrm{EL}_2(A) \hookrightarrow \mathrm{EL}_m(A)$ in any chosen pair of coordinates and multiply the resulting matrices for the commutator factors of c . $\square$

Whitehead
whitehead_commutator
whitehead_diagonal
MatrixDiagonalization
exists_elementary_whitehead

ElementaryGroup
elementaryGroup_
finitelyGenerated

Lemma 5.6. *Let A be a finitely generated unital ring and $r \geq 3$. Then $\text{EL}_r(A)$ is finitely generated: if A is generated as a unital ring by the finite set T , then $\text{EL}_r(A)$ is generated by the finitely many matrices $x_{ij}(1)$ and $x_{ij}(t)$, $t \in T$, $i \neq j$.*

Proof. Let E be the subgroup generated by the listed matrices, and let $A_0 \subseteq A$ be the set of a with $x_{ij}(a) \in E$ for all $i \neq j$. By the addition relation, A_0 is an additive subgroup containing 1 and T ; by the Steinberg relation with a third index $l \notin \{i, j\}$ (here $r \geq 3$ is used), $a, b \in A_0$ implies $ab \in A_0$, since $x_{ij}(ab) = [x_{il}(a), x_{lj}(b)]$ and $x_{il}(a), x_{lj}(b) \in E$. Hence A_0 is a unital subring containing T , so $A_0 = A$ and $E = \text{EL}_r(A)$. $\square$

formalized in part
FreeElementaryPropertyT
controlSet_isKazhdanPair
The arbitrary finitely
generated ring statement is
manuscript-only.

Theorem 5.7 (Ershov–Jaikin-Zapirain [9]). *Let A be a finitely generated associative unital ring and $r \geq 3$. Then $\text{EL}_r(A)$ has Kazhdan’s property (T).*

The theorem in [9] is stated for the universal finitely generated rings $\mathbb{Z}\langle x_1, \dots, x_d \rangle$; the general case follows since EL_r of a quotient ring is a quotient group of EL_r of the ring, elementary generators mapping to elementary generators, and property (T) passes to quotients.

Over a ring carrying a binary Leavitt family, the rank of an elementary group is immaterial: matrix self-similarity collapses all ranks at the level of the *elementary* groups themselves, with no K -theory and no GE input. This is the engine that will spread every property of one rank to all ranks.

LeavittRankEquivalence
rankSuccEquiv
rankSuccToProduct
rankSucc_propertyT_of_
rankSucc

Proposition 5.8 (Elementary rank equivalence). *Let A be a unital ring carrying a binary Leavitt family and let $n, m \geq 2$. Then there is an explicit group isomorphism*

$$\text{EL}_n(A) \cong \text{EL}_m(A),$$

the composite

$$\text{EL}_n(A) \cong \text{EL}_n(M_m(A)) = \text{EL}_{nm}(A) = \text{EL}_m(M_n(A)) \cong \text{EL}_m(A),$$

where the outer isomorphisms apply Θ_C^{-1} and $\Theta_{C'}$ of Proposition 5.4 to each matrix coefficient, and the middle equalities are block flattening. In particular finite generation, infinitude, property (T), and (non)soficity hold at one rank ≥ 2 if and only if they hold at every rank ≥ 2 .

Proof. Coefficient transport. A unital ring isomorphism $\varphi : B \rightarrow B'$ carries $x_{ij}(b)$ to $x_{ij}(\varphi b)$, hence induces an isomorphism $\text{EL}_p(B) \cong \text{EL}_p(B')$ for every p . Apply this with $\varphi = \Theta_{C_m}^{-1} : A \rightarrow M_m(A)$, which exists by Proposition 5.4.

Block flattening. Identify $M_p(M_q(A)) = M_{pq}(A)$ by the index pairing (i, c) , $1 \leq i \leq p$, $1 \leq c \leq q$. We claim $\text{EL}_p(M_q(A)) = \text{EL}_{pq}(A)$ as subgroups of $\text{GL}_{pq}(A)$, for every $p \geq 2$ and $q \geq 1$. One inclusion is direct: a block transvection $I + aE_{ij}$, $i \neq j$, $a \in M_q(A)$, equals the product $\prod_{c,d} x_{(i,c),(j,d)}(a_{cd})$ of q^2 commuting entry transvections. For the other inclusion, an entry transvection $x_{(i,c),(j,d)}(r)$ with $i \neq j$ is a block transvection with the single-entry coefficient rE_{cd} ; and one with $i = j$ (so $c \neq d$) is produced by the

Steinberg relation through any third block index $l \neq i$, which exists because $p \geq 2$:

$$x_{(i,c),(i,d)}(r) = [x_{(i,c),(l,c)}(r), x_{(l,c),(i,d)}(1)],$$

both factors being cross-block transvections, hence in $\text{EL}_p(M_q(A))$ by the previous sentence. Applying block flattening with $(p, q) = (n, m)$ and—after reindexing along the bijection $(i, c) \mapsto (c, i)$, which conjugates by a permutation matrix and matches entry transvections to entry transvections—with $(p, q) = (m, n)$, gives the middle equalities. The final claim holds because all four properties are isomorphism-invariant. $\square$

5.3. Characteristic-two block factorizations. The following two identities, valid in characteristic two, are the engine of the explicit spine.

Lemma 5.9. *Let A be a unital ring of characteristic 2 carrying a binary Leavitt family. Then, in $M_2(A)$:*

- (a) $x_{12}(s_1) x_{21}(t_1) x_{12}(s_1) = \begin{pmatrix} p_0 & s_1 \\ t_1 & 0 \end{pmatrix};$
- (b) $x_{21}(1 + t_0) x_{12}(1) x_{21}(1 + s_0) x_{12}(t_0) = \begin{pmatrix} s_0 & s_1 t_1 \\ 0 & t_0 \end{pmatrix}.$

*Leavitt
characteristicTwo_involution
characteristicTwo_compressor
compressor_factorization
The factorization is proved in
every characteristic.*

Proof. (a) Since $1 + p_1 = p_0$ in characteristic 2,

$$\begin{pmatrix} 1 & s_1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ t_1 & 1 \end{pmatrix} = \begin{pmatrix} 1 + s_1 t_1 & s_1 \\ t_1 & 1 \end{pmatrix} = \begin{pmatrix} p_0 & s_1 \\ t_1 & 1 \end{pmatrix},$$

and then

$$\begin{pmatrix} p_0 & s_1 \\ t_1 & 1 \end{pmatrix} \begin{pmatrix} 1 & s_1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} p_0 & p_0 s_1 + s_1 \\ t_1 & t_1 s_1 + 1 \end{pmatrix} = \begin{pmatrix} p_0 & s_1 \\ t_1 & 0 \end{pmatrix},$$

using $p_0 s_1 = s_0(t_0 s_1) = 0$ and $t_1 s_1 + 1 = 1 + 1 = 0$.

(b) Compute the product of the first two factors:

$$\begin{pmatrix} 1 & 0 \\ 1 + t_0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 + t_0 & t_0 \end{pmatrix}$$

(using $1 + t_0 + 1 = t_0$ in characteristic 2). Multiplying by the third factor,

$$\begin{pmatrix} 1 & 1 \\ 1 + t_0 & t_0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 1 + s_0 & 1 \end{pmatrix} = \begin{pmatrix} s_0 & 1 \\ 1 + t_0 + t_0 + t_0 s_0 & t_0 \end{pmatrix} = \begin{pmatrix} s_0 & 1 \\ 0 & t_0 \end{pmatrix},$$

where the $(1, 1)$ -entry is $1 + 1 + s_0 = s_0$ and the $(2, 1)$ -entry is $(1 + t_0) + t_0(1 + s_0) = 1 + t_0 + t_0 + t_0 s_0 = 1 + 1 = 0$, using $t_0 s_0 = 1$. Finally,

$$\begin{pmatrix} s_0 & 1 \\ 0 & t_0 \end{pmatrix} \begin{pmatrix} 1 & t_0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} s_0 & s_0 t_0 + 1 \\ 0 & t_0 \end{pmatrix} = \begin{pmatrix} s_0 & s_1 t_1 \\ 0 & t_0 \end{pmatrix},$$

since $s_0 t_0 + 1 = p_0 + 1 = p_1 = s_1 t_1$ in characteristic 2. $\square$

5.4. Thompson's group V in a Leavitt corner. Thompson's group V is the group of homeomorphisms of the boundary $\{0, 1\}^{\mathbb{N}}$ given by *tree tables*: a pair of ordered complete leaf sets $(\beta_1, \dots, \beta_\ell)$, $(\alpha_1, \dots, \alpha_\ell)$ with the same number ℓ of leaves, acting by the prefix substitutions $\beta_i x \mapsto \alpha_i x$; each such map carries the cylinder of β_i bijectively onto the cylinder of α_i and is a homeomorphism of the Cantor set. Two tables define the same element exactly when they have a common refinement, refinement being simultaneous subdivision of a source leaf and its target leaf. The group V is infinite, simple, and finitely presented [11, 6], and 2-generated [5].

Proposition 5.10 (Tree-table embedding). *Let A be a nontrivial unital ring carrying a binary Leavitt family. For $g \in V$ given by a table $((\beta_i), (\alpha_i))$, put*

$$(23) \quad U_g = \sum_{i=1}^{\ell} \alpha_i \beta_i^* \in A.$$

Then U_g is well defined (independent of the table), $U_g U_h = U_{gh}$, $U_1 = 1$, each U_g is a unit with $U_g^{-1} = U_{g^{-1}}$, and $g \mapsto U_g$ is an injective homomorphism $V \hookrightarrow A^\times$.

Proof. Refining a table replaces a summand $\alpha\beta^*$ by $\alpha 0 (\beta 0)^* + \alpha 1 (\beta 1)^* = \alpha(s_0 t_0 + s_1 t_1)\beta^* = \alpha\beta^*$; since any two tables of the same element admit a common refinement, U_g is well defined. For multiplicativity, choose tables of g and h such that the source leaf set of g equals the target leaf set of h (refine both accordingly); then

$$U_g U_h = \sum_{i,j} \alpha_i \beta_i^* \gamma_j \delta_j^* = \sum_i \alpha_i \delta_i^* = U_{gh},$$

using $\beta_i^* \gamma_j = \delta_{ij}$ for the common leaf set, and the composed table represents gh . Also $U_1 = \sum_i \alpha_i \alpha_i^* = 1$ by Lemma 5.3, so each U_g is a unit with inverse $U_{g^{-1}}$. For injectivity, suppose $U_g = 1$ with a table $((\beta_i), (\alpha_i))$. Multiplying $\sum_i \alpha_i \beta_i^* = 1$ on the right by β_j gives $s_{\alpha_j} = s_{\beta_j}$ in A for each j —the products $\beta_i^* \beta_j = \delta_{ij}$ collapse the sum. Equality of these elements forces equality of the binary words. If neither word is a prefix of the other, multiplying $s_{\alpha_j} = s_{\beta_j}$ on the left by $s_{\alpha_j}^*$ gives $1 = s_{\alpha_j}^* s_{\beta_j} = 0$, contradicting the nontriviality of A . If, say, $\beta_j = \alpha_j \gamma$ with γ nonempty, the same multiplication gives $1 = s_\gamma$; choosing $i \in \{0, 1\}$ different from the first letter of γ then gives $t_i = t_i s_\gamma = 0$, whence $1 = t_i s_i = 0$, again a contradiction. The case $\alpha_j = \beta_j \gamma$ is symmetric. Hence $\alpha_j = \beta_j$ as words for every j , and g is the identity substitution. This is the standard realization of the Higman–Thompson groups inside Leavitt algebras and their Cuntz-algebra completions; compare [16, 17]. $\square$

Lemma 5.11 (Corner embedding). *Let A be a nontrivial unital ring carrying a binary Leavitt family. For $g \in V$ put*

$$\rho(g) = p_0 + s_1 U_g t_1 \in A.$$

ThompsonVEmbedding
vEmbedding
vEmbedding_injective
isComplete_of_covers
ThompsonV
tableEquiv

LeavittCorner
cornerHom
cornerHom_s0
t0_cornerHom
t1_cornerHom_s1

Then $\rho : V \rightarrow A^\times$ is an injective group homomorphism, and for all $g \in V$:

$$t_1 \rho(g) s_1 = U_g, \quad \rho(g) s_0 = s_0, \quad t_0 \rho(g) = t_0.$$

Proof. All computations take place in the subring generated by the Leavitt family, using $t_1 s_0 = 0$, $t_0 s_1 = 0$, $t_1 s_1 = t_0 s_0 = 1$, $p_0 s_0 = s_0$, $t_0 p_0 = t_0$, $p_0 s_1 = 0$, $t_1 p_0 = 0$. Multiplicativity:

$$\begin{aligned} \rho(g) \rho(h) &= (p_0 + s_1 U_g t_1)(p_0 + s_1 U_h t_1) \\ &= p_0^2 + p_0 s_1 U_h t_1 + s_1 U_g t_1 p_0 + s_1 U_g t_1 s_1 U_h t_1 \\ &= p_0 + s_1 U_g U_h t_1 = \rho(gh), \end{aligned}$$

by Proposition 5.10. Also $\rho(1) = p_0 + p_1 = 1$, so $\rho(g) \in A^\times$ with inverse $\rho(g^{-1})$. The three displayed identities are immediate: $t_1 \rho(g) s_1 = t_1 p_0 s_1 + t_1 s_1 U_g t_1 s_1 = U_g$; $\rho(g) s_0 = p_0 s_0 + s_1 U_g t_1 s_0 = s_0$; $t_0 \rho(g) = t_0 p_0 + t_0 s_1 U_g t_1 = t_0$. Injectivity: $\rho(g) = 1$ forces $U_g = t_1 \rho(g) s_1 = t_1 s_1 = 1$, hence $g = 1$ by Proposition 5.10. $\square$

Thus ρ realizes V inside the corner $p_0 + s_1(\cdot)t_1$, acting as the identity on the complementary corner $s_0(\cdot)t_0$: this complementarity is the algebraic seed of the centralizer relation $[K, J] = 1$.

Proposition 5.12. *Thompson's group V is finitely generated and is not LEF. Moreover V is perfect.*

Proof. V is finitely presented and infinite [11, 6] and even 2-generated [5]. If V were LEF, then being finitely presented it would be residually finite by Lemma 2.8; but V is infinite and simple, so its only finite quotient is trivial and it is not residually finite. Perfectness: $[V, V]$ is a nontrivial normal subgroup of the simple group V (it is nontrivial since V is nonabelian, containing for instance a copy of every finite symmetric group permuting the leaves of a fixed tree), hence $[V, V] = V$. $\square$

ThompsonVWitness
thompsonV_not_isLEF
ThompsonWitness
not_isLEF_
cornerWitnessSubgroup
Scheme
not_isLEF_cornerSubgroup
Strengthened: proved for V
itself, with no Higman
presentation input.

6. THE COMPRESSION SCHEME IN ADJACENT RANKS

This section presents the algebraic heart of the paper in its natural generality. The matrix identities below are verifiable by finite computation from the two Leavitt relations, over an arbitrary ambient ring. Beyond them, the section uses only the facts about Thompson's group V established in Section 5 and the two membership hypotheses isolated in Theorem 6.7, the latter discharged explicitly in Section 7 and K -theoretically in Section 8.

Throughout, A is a nontrivial associative unital ring carrying a binary Leavitt family (s_0, s_1, t_0, t_1) in the sense of Definition 5.1—in particular A may be, but need not contain a copy of, the Leavitt algebra $L = L_k(1, 2)$ —and we retain the notation $p_0 = s_0 t_0$, $p_1 = s_1 t_1$. Fix $m \geq 2$ and put

$$n = m + 1, \quad G = \text{EL}_n(A), \quad \Gamma = \{\hat{\gamma} = \text{diag}(\gamma, 1) \mid \gamma \in \text{EL}_m(A)\} \leq G.$$

Indeed $\Gamma \leq G$: an elementary matrix in the first m coordinates is an elementary matrix of rank n . The group $\Gamma \cong \text{EL}_m(A)$ is infinite, since $a \mapsto x_{12}(a)$ is injective and A is infinite by Lemma 5.2.

We use the left comb (22) with n leaves:

$$(24) \quad \alpha_i = s_0^{i-1} s_1 \quad (1 \leq i \leq m), \quad \alpha_n = s_0^m,$$

a complete leaf set, so that $\alpha_i^* \alpha_j = \delta_{ij}$ and $\sum_{i=1}^n \alpha_i \alpha_i^* = 1$ by Lemma 5.3.

6.1. The comb compressor. Define the column $b \in A^{m \times 1}$, the row $c \in A^{1 \times m}$, and the matrices $u, u' \in M_n(A)$:

$$(25) \quad b = \begin{pmatrix} s_1 \alpha_1^* \\ \vdots \\ s_1 \alpha_m^* \end{pmatrix}, \quad c = (\alpha_1 t_1 \quad \cdots \quad \alpha_m t_1),$$

$$u = \begin{pmatrix} s_0 I_m & b \\ 0 & \alpha_n^* \end{pmatrix}, \quad u' = \begin{pmatrix} t_0 I_m & 0 \\ c & \alpha_n \end{pmatrix}.$$

Lemma 6.1. *The following identities hold in A :*

$$bc = p_1 I_m, \quad cb = 1 - \alpha_n \alpha_n^*, \quad t_0 b = 0, \quad cs_0 = 0, \quad b \alpha_n = 0, \quad \alpha_n^* c = 0.$$

Consequently $uu' = u'u = I_n$; thus $u \in \text{GL}_n(A)$ with $u^{-1} = u'$.

Proof. Entrywise: $(bc)_{ij} = s_1 \alpha_i^* \alpha_j t_1 = \delta_{ij} s_1 t_1 = \delta_{ij} p_1$. Next, $cb = \sum_{i=1}^m \alpha_i t_1 s_1 \alpha_i^* = \sum_{i=1}^m \alpha_i \alpha_i^* = 1 - \alpha_n \alpha_n^*$ by Lemma 5.3. The entries of $t_0 b$ are $t_0 s_1 \alpha_i^* = 0$; those of cs_0 are $\alpha_i t_1 s_0 = 0$; those of $b \alpha_n$ are $s_1 \alpha_i^* \alpha_n = 0$ and of $\alpha_n^* c$ are $\alpha_n^* \alpha_i t_1 = 0$, by orthogonality of distinct leaves. Then

$$uu' = \begin{pmatrix} s_0 t_0 I_m + bc & b \alpha_n \\ \alpha_n^* c & \alpha_n^* \alpha_n \end{pmatrix} = \begin{pmatrix} (p_0 + p_1) I_m & 0 \\ 0 & 1 \end{pmatrix} = I_n,$$

and likewise

$$u'u = \begin{pmatrix} t_0 s_0 I_m & t_0 b \\ cs_0 & cb + \alpha_n \alpha_n^* \end{pmatrix} = I_n. \quad \square$$

Proposition 6.2 (Compression). *For every $\gamma \in \text{EL}_m(A)$,*

$$(26) \quad u \hat{\gamma} u^{-1} = p_1 \widehat{I_m + s_0 \gamma t_0},$$

and on elementary generators $u \hat{x}_{ij}(a) u^{-1} = \hat{x}_{ij}(s_0 a t_0)$ for $i \neq j \leq m$. Consequently

$$(27) \quad K := u \Gamma u^{-1} = \langle \hat{x}_{ij}(s_0 a t_0) : a \in A, i \neq j \leq m \rangle \leq \Gamma,$$

and conjugation by u is an isomorphism $\Gamma \xrightarrow{\cong} K$; in particular $K \cong \text{EL}_m(A)$.

Proof. Block multiplication using Lemma 6.1:

$$u \hat{\gamma} u^{-1} = \begin{pmatrix} s_0 \gamma & b \\ 0 & \alpha_n^* \end{pmatrix} \begin{pmatrix} t_0 I_m & 0 \\ c & \alpha_n \end{pmatrix} = \begin{pmatrix} s_0 \gamma t_0 + bc & b \alpha_n \\ \alpha_n^* c & \alpha_n^* \alpha_n \end{pmatrix}$$

$$= \begin{pmatrix} s_0 \gamma t_0 + p_1 I_m & 0 \\ 0 & 1 \end{pmatrix}.$$

LeavittMatrixCompression
matrixCompressionHom

RankFourCompressors
compressor_conjugation
compressionEnd_injective

For $\gamma = x_{ij}(a) = I_m + aE_{ij}$, the block is $p_1I_m + s_0t_0I_m + s_0at_0E_{ij} = I_m + s_0at_0E_{ij} = x_{ij}(s_0at_0)$. Since Γ is generated by the $\hat{x}_{ij}(a)$ and conjugation is a group isomorphism onto its image, K is generated by the displayed elements, which lie in Γ . $\square$

The compressed copy K sits inside Γ with all matrix entries confined to the corner $s_0(\cdot)t_0 = p_0Ap_0$ of A ; the complementary corner p_1Ap_1 is untouched, and this is the room in which the centralizing subgroup J will live.

6.2. The involution z and generation. Let $z \in M_n(A)$ agree with the identity matrix outside the coordinates 1 and n , where it is

$$z|_{\{1,n\}} = \begin{pmatrix} p_0 & s_1 \\ t_1 & 0 \end{pmatrix}; \quad \text{that is,} \quad z = I_n + (p_0 - 1)E_{11} + s_1E_{1n} + t_1E_{n1} - E_{nn}. \quad (28)$$

Lemma 6.3. $z^2 = I_n$; in particular $z \in \text{GL}_n(A)$. Moreover z centralizes K .

Proof. On the coordinates $\{1, n\}$,

$$\begin{pmatrix} p_0 & s_1 \\ t_1 & 0 \end{pmatrix}^2 = \begin{pmatrix} p_0^2 + s_1t_1 & p_0s_1 \\ t_1p_0 & t_1s_1 \end{pmatrix} = \begin{pmatrix} p_0 + p_1 & 0 \\ 0 & 1 \end{pmatrix} = I_2,$$

using $p_0s_1 = 0$, $t_1p_0 = 0$, $t_1s_1 = 1$; the remaining coordinates are fixed. For the centralizing statement, put $r = s_0at_0$ and note

$$p_0r = r = rp_0, \quad t_1r = 0, \quad rs_1 = 0.$$

Let $i \neq j \leq m$. If $1 \notin \{i, j\}$, then z acts as the identity on the coordinates i, j and $z\hat{x}_{ij}(r)z = \hat{x}_{ij}(r)$. If $i = 1$: the matrix rE_{1j} satisfies $z(rE_{1j})z = (p_0rE_{1j} + t_1rE_{nj})z = rE_{1j}z = rE_{1j}$, since the j -th row of z is $e_j^\top$ for $2 \leq j \leq m$. If $j = 1$: $z(rE_{i1})z = rE_{i1}z = rp_0E_{i1} + rs_1E_{in} = rE_{i1}$. In each case $z\hat{x}_{ij}(r)z^{-1} = \hat{x}_{ij}(r)$, and these elements generate K by (27). $\square$

Define the second compressor

$$(29) \quad v = zu.$$

Then, by Proposition 6.2 and Lemma 6.3,

$$(30) \quad v\Gamma v^{-1} = z(u\Gamma u^{-1})z^{-1} = zKz^{-1} = K = u\Gamma u^{-1} :$$

the two compressors have one common image. The quotient $vu^{-1} = z$ is precisely what generates the missing last row and column:

Proposition 6.4 (Generation). *The subgroup of $\text{GL}_n(A)$ generated by Γ and z contains $\text{EL}_n(A)$. Consequently, if u and z lie in $\text{EL}_n(A)$, then*

$$\text{EL}_n(A) = \langle \Gamma, z \rangle = \langle \Gamma, u, v \rangle.$$

Proof. For $2 \leq j \leq m$ and $a \in A$, we claim

$$z\hat{x}_{1j}(s_1a)z = x_{nj}(a), \quad z\hat{x}_{j1}(at_1)z = x_{jn}(a).$$

For the first: $z(s_1aE_{1j})z = (p_0s_1aE_{1j} + t_1s_1aE_{nj})z = aE_{nj}z = aE_{nj}$, using $p_0s_1 = 0$, $t_1s_1 = 1$, and that the j -th row of z is $e_j^\top$. For the second:

Leavitt
z_sq
RankFourCompressors
involution_sq
DiagonalCornerCompression
matrixCompression_commutates_
firstDiagonalCorner

RankFourCompressors
coreEmbedding_compressorSet_
generate

$z(at_1E_{j1})z = at_1E_{j1}z = at_1p_0E_{j1} + at_1s_1E_{jn} = aE_{jn}$, using $t_1p_0 = 0$, $t_1s_1 = 1$, and that the j -th column of z is e_j . Thus $\langle \Gamma, z \rangle$ contains $x_{nj}(a)$ and $x_{jn}(a)$ for all $2 \leq j \leq m$ and $a \in A$, as well as all $\hat{x}_{ij}(a) = x_{ij}(a)$ with $i \neq j \leq m$. The remaining elementary matrices are supplied by the Steinberg relation, using $m \geq 2$ so that the coordinate $2 \notin \{1, n\}$ exists:

$$x_{1n}(a) = [x_{12}(a), x_{2n}(1)], \quad x_{n1}(a) = [x_{n2}(a), x_{21}(1)].$$

Hence $\langle \Gamma, z \rangle \supseteq \text{EL}_n(A)$. If moreover $u, z \in \text{EL}_n(A)$, then $\langle \Gamma, z \rangle \subseteq \text{EL}_n(A)$, giving the first equality; and since $z = vu^{-1}$, also $\langle \Gamma, u, v \rangle \supseteq \langle \Gamma, z \rangle = \text{EL}_n(A) \supseteq \langle \Gamma, u, v \rangle$. $\square$

6.3. The commuting corner copy of V . For $g \in V$, recall $\rho(g) = p_0 + s_1U_g t_1 \in A^\times$ from Lemma 5.11, and define

$$j(g) = \text{diag}(\rho(g), 1, \dots, 1) \in \text{GL}_n(A), \quad J = \{j(g) \mid g \in V\}.$$

Proposition 6.5. *For every $g \in V$, the matrix $\text{diag}(\rho(g), 1, \dots, 1) \in \text{GL}_m(A)$ lies in $\text{EL}_m(A)$. Hence $J \leq \Gamma$, the map $j : V \rightarrow J$ is an isomorphism, and J is a finitely generated subgroup of Γ that is not LEF.*

Proof. By Proposition 5.12, V is perfect, so $g = \prod_i [a_i, b_i]$ for some $a_i, b_i \in V$. Since ρ is a homomorphism (Lemma 5.11), $\rho(g) = \prod_i [\rho(a_i), \rho(b_i)]$ is a product of commutators of units of A . The Whitehead commutator lemma (Lemma 5.5, with $m \geq 2$) places $\text{diag}(\rho(g), 1, \dots, 1)$ in $\text{EL}_m(A)$. Thus $j(g) \in \Gamma$. The map j is a homomorphism because ρ is, and it is injective because ρ is; so $J \cong V$, which is finitely generated and not LEF by Proposition 5.12. $\square$

Proposition 6.5 holds for every unital ring A containing the Leavitt family: the corner copy of V is elementary with no hypothesis on A beyond the Leavitt family itself.

Proposition 6.6. $[K, J] = 1$ and $K \cap J = \{1\}$. Consequently $KJ \cong K \times J$.

Proof. Centralizing. Conjugating an elementary matrix by a diagonal matrix multiplies its coefficient: $j(g) \hat{x}_{il}(r) j(g)^{-1} = \hat{x}_{il}(\rho(g)^{[i=1]} r \rho(g^{-1})^{[l=1]})$, where the exponent $[i = 1]$ is 1 if $i = 1$ and 0 otherwise. For $r = s_0at_0$, Lemma 5.11 gives $\rho(g)r = \rho(g)s_0at_0 = s_0at_0 = r$ and $r\rho(g^{-1}) = s_0at_0\rho(g^{-1}) = s_0at_0 = r$. Hence $j(g)$ commutes with every generator $\hat{x}_{il}(s_0at_0)$ of K from (27).

Trivial intersection. Suppose $j(g) \in K$. By (26), every element of K has the form $p_1 \widehat{I_m} + s_0\gamma t_0$ with $\gamma \in \text{EL}_m(A)$; comparing $(1, 1)$ -entries, $\rho(g) = p_1 + s_0\gamma_{11}t_0$. Sandwiching between t_1 and s_1 : on one side, $t_1(\rho(g) - 1)s_1 = t_1(s_1U_g t_1 - p_1)s_1 = U_g - 1$, using $t_1p_0 = 0$; on the other, $t_1(p_1 + s_0\gamma_{11}t_0 - 1)s_1 = t_1(s_0\gamma_{11}t_0 - p_0)s_1 = 0$, using $t_1s_0 = 0$ and $t_1p_0 = 0$. Hence $U_g = 1$, so $g = 1$ by Proposition 5.10, and $j(g) = 1$.

Since K and J commute and intersect trivially, the multiplication map $K \times J \rightarrow KJ$ is an isomorphism. $\square$

UniversalRankFour
witnessEmbedding
DiagonalElementary
firstDiagonalUnit_mem_of_
mem_commutator

Scheme
commute_compressed_corner
compressed_inf_corner

6.4. The Leavitt-corner theorem.

Theorem 6.7 (Leavitt-corner theorem). *Let A be a nontrivial associative unital ring carrying a binary Leavitt family, let $m \geq 2$, and put $n = m + 1$. Let $u, z \in \mathrm{GL}_n(A)$ be the matrices (25) and (28). Assume:*

- (i) $\mathrm{EL}_m(A)$ and $\mathrm{EL}_n(A)$ have property (T);
- (ii) $u, z \in \mathrm{EL}_n(A)$; this holds, for instance, whenever $\mathrm{GL}_n(A) = \mathrm{EL}_n(A)$.

Then $\mathrm{EL}_n(A)$ is not sofic.

Proof. Put $G = \mathrm{EL}_n(A)$ and let $\Gamma, u, z, v = zu, K, J$ be as above. By hypothesis (i), G and $\Gamma \cong \mathrm{EL}_m(A)$ are property-(T) groups, hence finitely generated, and Γ is infinite. By hypothesis (ii) and Proposition 6.4, $G = \langle \Gamma, u, v \rangle = \langle \Gamma, Q \rangle$ with $Q = \{u, v\} \subseteq G$. By Proposition 6.2 and (30),

$$u\Gamma u^{-1} = v\Gamma v^{-1} = K \leq \Gamma,$$

so both elements of Q conjugate Γ into itself. Fix $q_0 = u$, so $K = q_0\Gamma q_0^{-1}$. By Propositions 6.5 and 6.6, the subgroup $J \leq \Gamma$ is finitely generated with $[K, J] = 1$ and $K \cap J = \{1\}$, and $J \cong V$ is not LEF. If G were sofic, Corollary 4.2 (Theorem D) would force J to be LEF, a contradiction. $\square$

Corollary 6.8. *Let A be a nontrivial finitely generated unital ring carrying a binary Leavitt family, and let $m \geq 3$, $n = m + 1$. If $u, z \in \mathrm{EL}_n(A)$ —for instance, if $\mathrm{GL}_n(A) = \mathrm{EL}_n(A)$ —then $\mathrm{EL}_n(A)$ is an infinite, finitely generated, property-(T) group that is not sofic.*

Proof. Theorem 5.7 supplies hypothesis (i) of Theorem 6.7 for both ranks $m, n \geq 3$, so $\mathrm{EL}_n(A)$ is not sofic and has property (T). It is finitely generated by Lemma 5.6, and infinite since it contains $J \cong V$. $\square$

Remark 6.9 (The trust surface of the scheme). The matrix identities of this section are verifiable by finite computation from the two Leavitt relations; beyond them, the section inputs only the Whitehead lemma, the finite generation, perfectness, and non-LEF property of V established in Section 5, and hypotheses (i) and (ii) of Theorem 6.7. The corner copy of V is elementary for every ambient ring: Proposition 6.5 uses only the perfectness of V and the Whitehead lemma. The entire membership burden therefore falls on hypothesis (ii), which concerns exactly two matrices, u and z : Section 7 discharges it by explicit elementary factorizations in characteristic two, and Section 8 discharges it over every finite field via $K_1(L) = 0$ and the GE property, with the K_1 input available in elementary, formalized form (Remark 8.5).

7. THE EXPLICIT REALIZATION: NO K -THEORY

In this section $k = \mathbb{F}_2$ and $R = L_{\mathbb{F}_2}(1, 2)$; the matrix identities and the argument extend without change to any finite field of characteristic two. Fix $m \geq 3$, put $n = m + 1$, and let $G = \mathrm{EL}_n(R)$, Γ, u, z, v, K, J be as in Section 6 with $A = R$. We now exhibit the two matrices u and z as explicit

CompressionSetup
CompressionSetup
UniversalCompressionSetup
compressionSetup

CriterionAssembly
not_isSofic_of_not_isLEF

products of elementary matrices, discharging hypothesis (ii) of Theorem 6.7 with no K -theoretic input.

Proposition 7.1 (Explicit elementary factorization of the compressor). *For $1 \leq i \leq m$, put*

$$U_i = x_{ni}(1 + t_0) x_{in}(1) x_{ni}(1 + s_0) x_{in}(t_0) \in \text{EL}_n(R).$$

Then

$$u = U_m U_{m-1} \cdots U_1.$$

In particular $u \in \text{EL}_n(R)$, as a product of $4m$ elementary matrices.

Proof. By Lemma 5.9(b), applied in the pair of coordinates (i, n) , the matrix U_i agrees with the identity outside the coordinates i and n and equals $\begin{pmatrix} s_0 & s_1 t_1 \\ 0 & t_0 \end{pmatrix}$ on them; explicitly,

$$U_i = I_n + (s_0 + 1)E_{ii} + s_1 t_1 E_{in} + (t_0 + 1)E_{nn},$$

signs being immaterial in characteristic two. Put $P_k = U_k U_{k-1} \cdots U_1$. We claim, by induction on k , that

$$P_k = \sum_{i \leq k} (s_0 E_{ii} + s_1 t_1 t_0^{i-1} E_{in}) + \sum_{k < i \leq m} E_{ii} + t_0^k E_{nn}.$$

For $k = 1$ this is the displayed form of U_1 . For the inductive step, compute the rows of $P_{k+1} = U_{k+1} P_k$. For $i \notin \{k+1, n\}$, the i -th row of U_{k+1} is $e_i^\top$, so the i -th row of P_{k+1} equals that of P_k . The $(k+1)$ -st row of U_{k+1} is $s_0 e_{k+1}^\top + s_1 t_1 e_n^\top$; since the $(k+1)$ -st row of P_k is $e_{k+1}^\top$ and the n -th row of P_k is $t_0^k e_n^\top$, the $(k+1)$ -st row of P_{k+1} is $s_0 e_{k+1}^\top + s_1 t_1 t_0^k e_n^\top$. The n -th row of U_{k+1} is $t_0 e_n^\top$, so the n -th row of P_{k+1} is $t_0^{k+1} e_n^\top$. This is the claimed form for $k+1$.

Finally, for the left comb (24), $s_1 t_1 t_0^{i-1} = s_1 (s_0^{i-1} s_1)^* = s_1 \alpha_i^*$ for $1 \leq i \leq m$, and $t_0^m = \alpha_n^*$. Hence

$$P_m = \sum_{i=1}^m (s_0 E_{ii} + s_1 \alpha_i^* E_{in}) + \alpha_n^* E_{nn} = \begin{pmatrix} s_0 I_m & b \\ 0 & \alpha_n^* \end{pmatrix} = u,$$

by (25). □

Lemma 7.2 (Explicit elementary factorization of the involution).

$$z = x_{1n}(s_1) x_{n1}(t_1) x_{1n}(s_1) \in \text{EL}_n(R).$$

Proof. Lemma 5.9(a), applied in the pair of coordinates $(1, n)$, produces exactly the matrix z of (28). □

Theorem 7.3 (The explicit spine). *For every $m \geq 3$, the group $\text{EL}_{m+1}(R)$ over $R = L_{\mathbb{F}_2}(1, 2)$ is infinite, finitely generated, has property (T), and is not sofic.*

Proof. The ring $R = L_{\mathbb{F}_2}(1, 2)$ is finitely generated (by s_0, s_1, t_0, t_1 over the prime field). Proposition 7.1 and Lemma 7.2 verify $u, z \in \text{EL}_{m+1}(R)$, and Corollary 6.8 applies. □

RankFourCompressors

compressor

compressorMatrix

compressorPiece

Explicit elementary words, in every characteristic.

RankFourCompressors

involution

involutionMatrix

involution_sq

Explicit elementary words, in every characteristic.

MainResults

universalLeavittEL4_not_

isSofic

universalLeavitt_profile

Proof of Theorem A. Theorem 7.3 gives every $m \geq 3$; the case $m = 3$ is the group $G = \text{EL}_4(R)$. For $m \in \{1, 2\}$, the elementary rank equivalence of Proposition 5.8 is an explicit isomorphism $\text{EL}_{m+1}(R) \cong \text{EL}_4(R)$, and all four properties transport along it; in particular $\text{EL}_2(R)$ and $\text{EL}_3(R)$ have property (T) although Theorem 5.7 says nothing at rank two. The two auxiliary matrices are exhibited as explicit products of elementary matrices: u by Proposition 7.1 and z (hence $v = zu$) by Lemma 7.2. The remaining memberships, of the diagonal matrices $j(g)$, $g \in V$, follow from the explicit Whitehead identity of Lemma 5.5 together with the perfectness of V (Proposition 6.5); once a commutator expression of g in V is chosen, the Whitehead identity converts it into an explicit elementary word. No K -theory is used. $\square$

Remark 7.4 (Scope of the explicit spine). The factorizations of this section are identities in characteristic two and hold over any field of characteristic two; finiteness of the field is used only through finite generation of the ring in Theorem 5.7. Thus $\text{EL}_{m+1}(L_k(1, 2))$ is infinite, finitely generated, Kazhdan, and nonsfic, with all memberships explicit, for every finite field k of characteristic two and every $m \geq 3$ —and then for every $m \geq 1$ by the elementary rank equivalence (Proposition 5.8), still with no K -theory. For rank $n = 3$ (that is, $m = 2$), hypothesis (i) of Theorem 6.7 for EL_2 is not supplied by Theorem 5.7; it is exactly here that the rank equivalence substitutes. Section 8 reaches the remaining groups of Theorem B—the unit group and every GL_r , over every finite field—through self-similarity.

8. RANK TWO AND THE FULL PANORAMA

The compression–centralizer mechanism closes already in rank two, over a smaller group Γ : the full diagonal unit group. Combined with matrix self-similarity and the vanishing of K_1 (Proposition 8.3, with the elementary proof of Remark 8.5), this yields Theorem B.

8.1. The two-by-two realization.

Theorem 8.1 (Two-by-two Leavitt-corner theorem). *Let A be a nontrivial associative unital ring carrying a binary Leavitt family. Suppose that $A^\times$ and $\text{EL}_2(A)$ have property (T) and that $\text{GL}_2(A) = \text{EL}_2(A)$. Then $\text{EL}_2(A)$ is not sofic.*

*RankTwoCompression
rankTwo_not_isSofic
rankTwoInvolution_conj_
upperNil
rankTwoInvolution_conj_
lowerNil*

Proof. Put $G = \text{EL}_2(A) = \text{GL}_2(A)$ and

$$\Gamma = \{\text{diag}(a, 1) \mid a \in A^\times\} \leq G.$$

Then $\Gamma \cong A^\times$ is a property-(T) group, hence finitely generated, and it is infinite because $\rho : V \hookrightarrow A^\times$ is injective (Lemma 5.11).

The compressor. Put

$$u = \begin{pmatrix} s_0 & p_1 \\ 0 & t_0 \end{pmatrix}, \quad u' = \begin{pmatrix} t_0 & 0 \\ p_1 & s_0 \end{pmatrix}.$$

Using $p_1s_0 = s_1(t_1s_0) = 0$, $t_0p_1 = (t_0s_1)t_1 = 0$, $t_0s_0 = 1$, and $p_0 + p_1 = 1$:

$$uu' = \begin{pmatrix} s_0t_0 + p_1 & p_1s_0 \\ t_0p_1 & t_0s_0 \end{pmatrix} = I_2, \quad u'u = \begin{pmatrix} t_0s_0 & t_0p_1 \\ p_1s_0 & p_1 + s_0t_0 \end{pmatrix} = I_2,$$

so $u \in \mathrm{GL}_2(A) = G$ with $u^{-1} = u'$. For $a \in A$, put

$$\kappa(a) = p_1 + s_0at_0.$$

Then $\kappa(1) = 1$ and $\kappa(a)\kappa(a') = p_1 + s_0aa't_0 = \kappa(aa')$ by the same identities, so κ is unital and multiplicative and restricts to a group homomorphism $A^\times \rightarrow A^\times$, injective because $t_0\kappa(a)s_0 = t_0p_1s_0 + t_0s_0at_0s_0 = a$; and

$$\begin{aligned} u \operatorname{diag}(a, 1) u^{-1} &= \begin{pmatrix} s_0a & p_1 \\ 0 & t_0 \end{pmatrix} \begin{pmatrix} t_0 & 0 \\ p_1 & s_0 \end{pmatrix} \\ &= \begin{pmatrix} s_0at_0 + p_1 & p_1s_0 \\ t_0p_1 & t_0s_0 \end{pmatrix} = \operatorname{diag}(\kappa(a), 1). \end{aligned}$$

Hence

$$K := u\Gamma u^{-1} = \{\operatorname{diag}(\kappa(a), 1) \mid a \in A^\times\} \leq \Gamma.$$

The involution. Put

$$z = \begin{pmatrix} p_0 & s_1 \\ t_1 & 0 \end{pmatrix}, \quad \text{so} \quad z^2 = \begin{pmatrix} p_0^2 + s_1t_1 & p_0s_1 \\ t_1p_0 & t_1s_1 \end{pmatrix} = I_2,$$

using $p_0s_1 = 0$, $t_1p_0 = 0$, $t_1s_1 = 1$; thus $z \in \mathrm{GL}_2(A) = G$ and $z^{-1} = z$. For any $c \in A$,

$$(31) \quad z \operatorname{diag}(c, 1) z = \begin{pmatrix} p_0cp_0 + p_1 & p_0cs_1 \\ t_1cp_0 & t_1cs_1 \end{pmatrix}.$$

The element $\kappa(a)$ satisfies $p_0\kappa(a) = \kappa(a)p_0 = s_0at_0$, $t_1\kappa(a) = t_1$, and $\kappa(a)s_1 = s_1$; substituting into (31) gives $z \operatorname{diag}(\kappa(a), 1) z = \operatorname{diag}(s_0at_0 + p_1, 1) = \operatorname{diag}(\kappa(a), 1)$. Hence z centralizes K , and $v := zu \in G$ satisfies $v\Gamma v^{-1} = zKz^{-1} = K$.

Generation. For any $a \in A$, the elements $1 + p_0at_1$ and $1 + s_1ap_0$ are units of A , since $(p_0at_1)^2 = 0$ and $(s_1ap_0)^2 = 0$ (using $t_1p_0 = 0$ and $p_0s_1 = 0$); so the diagonal matrices $\operatorname{diag}(1 + p_0at_1, 1)$ and $\operatorname{diag}(1 + s_1ap_0, 1)$ lie in Γ . Substituting $c = 1 + p_0at_1$ and $c = 1 + s_1ap_0$ into (31) and using $t_1p_0 = 0$, $p_0s_1 = 0$, $t_1s_1 = 1$, $p_0^2 = p_0$:

$$(32) \quad z \operatorname{diag}(1 + p_0at_1, 1) z = \begin{pmatrix} 1 & p_0a \\ 0 & 1 \end{pmatrix} = x_{12}(p_0a),$$

$$(33) \quad z \operatorname{diag}(1 + s_1ap_0, 1) z = \begin{pmatrix} 1 & 0 \\ ap_0 & 1 \end{pmatrix} = x_{21}(ap_0).$$

Next, the *corner swap*

$$d = s_1t_0 + s_0t_1 \in A^\times, \quad d^2 = s_1t_1 + s_0t_0 = 1,$$

gives $D = \text{diag}(d, 1) \in \Gamma$ with $D^{-1} = D$. Conjugation by D multiplies an upper coefficient on the left by d and a lower coefficient on the right by d ; since $d p_0 s_0 t_1 = d s_0 t_1 = p_1$ and $s_1 t_0 p_0 d = s_1 t_0 d = p_1$,

$$D x_{12}(p_0(s_0 t_1 a)) D = x_{12}(p_1 a), \quad D x_{21}((a s_1 t_0) p_0) D = x_{21}(a p_1),$$

with the inner matrices supplied by (32) and (33). Finally

$$x_{12}(a) = x_{12}(p_0 a) x_{12}(p_1 a), \quad x_{21}(a) = x_{21}(a p_0) x_{21}(a p_1),$$

so $\langle \Gamma, z \rangle$ contains every elementary matrix; hence $\langle \Gamma, z \rangle = G$, and since $z = v u^{-1}$,

$$G = \langle \Gamma, u, v \rangle, \quad u \Gamma u^{-1} = v \Gamma v^{-1} = K \leq \Gamma.$$

The commuting corner. Put $J = \{\text{diag}(\rho(g), 1) \mid g \in V\} \leq \Gamma$; membership is automatic here, since $\rho(g) \in A^\times$. By Lemma 5.11 and Proposition 5.12, $J \cong V$ is finitely generated and not LEF. The subgroups K and J commute:

$$\begin{aligned} \kappa(a) \rho(g) &= (p_1 + s_0 a t_0)(p_0 + s_1 U_g t_1) = s_1 U_g t_1 + s_0 a t_0 \\ &= (p_0 + s_1 U_g t_1)(p_1 + s_0 a t_0) = \rho(g) \kappa(a), \end{aligned}$$

using $p_1 s_1 = s_1$, $t_1 p_1 = t_1$, $t_0 p_0 = t_0$, $p_0 s_0 = s_0$, $t_0 s_1 = 0$, $t_1 s_0 = 0$. And $K \cap J$ is trivial: if $\kappa(a) = \rho(g)$, then sandwiching between t_0 and s_0 gives $a = t_0 \kappa(a) s_0 = t_0 \rho(g) s_0 = 1$, so $\rho(g) = \kappa(1) = 1$ and $g = 1$ by injectivity of ρ .

Conclusion. The data $G, \Gamma, Q = \{u, v\}$, $q_0 = u$, K, J satisfy all hypotheses of Corollary 4.2: G and Γ are finitely generated Kazhdan groups, Γ is infinite, $G = \langle \Gamma, Q \rangle$, each $q \in Q$ conjugates Γ into itself with $q_0 \Gamma q_0^{-1} = K$, and $J \leq \Gamma$ is finitely generated with $[K, J] = 1$ and $K \cap J = \{1\}$. If G were sofic, $J \cong V$ would be LEF, contradicting Proposition 5.12. $\square$

8.2. K_1 vanishes and $\text{GL} = \text{EL}$. We isolate the ring-theoretic external input in the exact form used.

Theorem 8.2 (Ara–Goodearl–Pardo [3]). *Let R be a purely infinite simple unital ring. Then:*

- (a) *R is a GE-ring: for every $r \geq 1$, every matrix in $\text{GL}_r(R)$ is a product of elementary matrices and a matrix $\text{diag}(a, 1, \dots, 1)$ with $a \in R^\times$;*
- (b) *the natural map $R^\times = \text{GL}_1(R) \rightarrow K_1(R)$ is surjective and induces an isomorphism $R^\times / [R^\times, R^\times] \cong K_1(R)$.*

Proposition 8.3. *Let k be any field and $L = L_k(1, 2)$. Then:*

- (i) $K_1(L) = 0$;
- (ii) *the unit group $L^\times$ is perfect;*
- (iii) $\text{GL}_r(L) = \text{EL}_r(L)$ *for every $r \geq 2$.*

Proof. Parts (i) and (iii) are proved, elementarily and in full, in Appendix A: Theorem A.2 establishes $K_1(L) = 0$ in Whitehead form—every unit u has $\text{diag}(u, 1) \in \text{EL}_2(L)$ —and Corollary A.23 deduces $\text{GL}_r(L) = \text{EL}_r(L)$ for every $r \geq 2$ via the elementary diagonalization of Lemma A.4. The only input of the appendix beyond the Leavitt relations is strong division (Lemma A.3),

formalized in part
 MatrixDiagonalization
 exists_elementary_mul_diag
 binaryLeavitt_exists_
 elementary_mul_diag
 LeavittSimplicity
 exists_mul_mul_eq_one
 The rank-two GE form is
 formalized in part (b)
 remains a cited input.
 ReimLoopDischarge
 glTwo_eq_elementary_holds
 glFour_eq_elementary_holds
 stableUnits_eq_top_holds
 $K_1(L) = 0$ in Whitehead
 form is unconditional; the
 $\text{GL} = \text{EL}$ collapse is
 formalized at ranks two and
 four.

which holds because L is purely infinite simple: the graph criteria of [1] are immediate for the rose R_2 with one vertex and two loops—the single vertex admits no nontrivial hereditary saturated subset, every cycle has an exit, and the vertex lies on a cycle.

For (ii): Theorem 8.2(b) identifies $K_1(L)$ with $L^\times/[L^\times, L^\times]$, so (i) gives $L^\times = [L^\times, L^\times]$.

For an independent confirmation of (i), the K -theory of Leavitt path algebras [2] gives the exact sequence

$$K_1(k) \xrightarrow{1-N^\top} K_1(k) \longrightarrow K_1(L) \longrightarrow K_0(k) \xrightarrow{1-N^\top} K_0(k),$$

where N is the adjacency matrix—here the 1×1 matrix (2), so $1 - N^\top = -1$, an isomorphism on both $K_0(k) \cong \mathbb{Z}$ and $K_1(k) \cong k^\times$, whence $K_1(L) = 0$. $\square$

Remark 8.4. For an independent recent account of generation phenomena for unit groups and matrix groups over Leavitt algebras, consistent with Proposition 8.3 and with the self-similarity of Proposition 5.4, see [12].

Remark 8.5 (An elementary route to $K_1(L) = 0$). The classical computation of $K_1(L)$ goes through the localization sequence of [2]. Appendix A replaces this input by a complete elementary proof—a Gaussian elimination over the binary tree, in the spirit of a Wiener–Hopf factorization whose “symbol” is a matrix pencil with two shift variables. In outline: a width reduction narrows every unit, modulo the Whitehead class group H of Definition A.1, to one whose value lies in the degree window $[-1, 1]$; at a sufficiently deep leaf set such a narrow unit becomes a matrix pencil with scalar coefficient matrices (A_0, A_1, C, B_0, B_1) over the five-dimensional space $kt_0 \oplus kt_1 \oplus k \oplus ks_0 \oplus ks_1$, indexed by a pair of complete leaf sets; and a single loop with two exits eliminates the pencil, splitting one column per pass (the index shift of the Wiener–Hopf analogy) until either a reshape by two tree tables lands the value in a one-sided degree window, or the stacked matrix $[B_0; B_1]$ acquires a scalar left inverse and the *inverse* unit lands there instead; one-sided windows die by a block-move induction whose base is the keystone Lemma A.16. Every step is carried out in Appendix A and formalized in the accompanying Lean development, so the K_1 -theoretic input of [2] is eliminated from the trust surface of Theorem B; the citation is retained as an independent confirmation.

Remark 8.6 (Effective Wiener–Hopf factorization over the binary tree). The elimination of Remark 8.5, written out in Appendix A, is an algorithm, and it yields a factorization theorem of independent interest. Call a unit *nonpositive* (resp. *nonnegative*) if its value lies in the span of monomials $s_\alpha t_\beta$ with $|\alpha| \leq |\beta|$ (resp. $\geq$); these are the two “triangular” classes of the tree-coded Wiener–Hopf analogy, and each lies in H by an explicit block-move induction. The algorithm shows: *every unit of $L = L_k(1, 2)$ factors as a central scalar times a product of tree-table units (Thompson’s group V), code-scalar units (elements of $\mathrm{GL}_n(k)$ transported along a complete prefix code), and a single nonpositive unit conjugated by two tree tables.* All factors are

RefineLoopDischarge
K1_trivial
narrowReduction_holds
scalarReduction_holds

formalized in part
RefineLoopDischarge
narrowReduction_holds
The elimination behind it is
formalized; the complexity
bounds are manuscript-only.

effectively computable: given u supported on monomials of depth at most d , the width reduction and the pencil loop run at codes with at most 2^{d+2} words, each loop step is Gaussian elimination over k on matrices of that size, and the loop executes at most 2^{d+1} times—so the factorization, and with it an explicit factorization of any $\text{diag}(u, 1, \dots, 1)$ into elementary matrices, is computed in time polynomial in 2^d and in the number of monomials of u . In the classical Wiener–Hopf picture the loop plays the role of index normalization: each refinement is the index shift that trades a constant diagonal for a shift atom, and the two exits are the two triangular factorizations. We are not aware of a prior effective factorization—or of any elementary proof of $K_1(L_k(1, 2)) = 0$ —in the literature; the localization-sequence computation of [2] is neither elementary nor effective.

8.3. Finite fields: all ranks and the unit group.

Theorem 8.7. *Let k be a finite field and $L = L_k(1, 2)$. Then $\text{EL}_2(L) = \text{GL}_2(L)$ is infinite, finitely generated, has property (T), and is not sofic.*

RankTwoCompression
rankTwo_not_isSofic

Proof. The ring L is finitely generated, by s_0, s_1, t_0, t_1 and the finitely many scalars. By Theorem 5.7 and Lemma 5.6, $\text{EL}_4(L)$ is a finitely generated Kazhdan group; by Proposition 8.3(iii), $\text{EL}_4(L) = \text{GL}_4(L)$; and by Proposition 5.4 with the left comb $\mathcal{C}_4$, $\text{GL}_4(L) \cong L^\times$. Hence $L^\times$ is a finitely generated Kazhdan group. Similarly, $\text{GL}_2(L) = \text{EL}_2(L)$ by Proposition 8.3(iii), and $\text{GL}_2(L) \cong L^\times$ by Proposition 5.4 with $\mathcal{C}_2 = (s_1, s_0)$; so $\text{EL}_2(L)$ is Kazhdan as well. Theorem 8.1 with $A = L$ now shows that $\text{EL}_2(L)$ is not sofic. It is infinite, containing $\text{diag}(\rho(V), 1) \cong V$, and finitely generated, being Kazhdan. $\square$

Theorem 8.8 (Panorama). *Let k be a finite field and $L = L_k(1, 2)$. Then the groups*

$$L^\times = \text{GL}_1(L), \quad \text{GL}_r(L) \ (r \geq 1), \quad \text{EL}_r(L) \ (r \geq 2)$$

are mutually isomorphic—explicitly, $\Theta_{\mathcal{C}} : \text{GL}_r(L) \rightarrow L^\times$ is an isomorphism for every ordered complete leaf set $\mathcal{C}$ of size r , and $\text{GL}_r(L) = \text{EL}_r(L)$ for $r \geq 2$ —and each is an infinite, finitely generated, property-(T) group that is not sofic.

MainResults
binaryLeavitt_finiteField_profile
binaryLeavittUnits_not_isSofic
binaryLeavittGL_not_isSofic
universalLeavittEL3_not_isSofic
Proved over every finite field via self-similarity, without K_1 or GE.

Proof. The isomorphisms are Proposition 5.4, and the equalities $\text{GL}_r(L) = \text{EL}_r(L)$ are Proposition 8.3(iii). Each of the listed groups is therefore isomorphic to $\text{EL}_2(L)$, which is infinite, finitely generated, Kazhdan, and nonsolic by Theorem 8.7; all four properties are isomorphism-invariant. $\square$

Proof of Theorem B. Part (i) and part (ii) are exactly Theorem 8.8, with the explicit isomorphism (2) given by Proposition 5.4. $\square$

Remark 8.9 (Two routes to rank four). For a finite field of characteristic two and rank $m + 1 \geq 4$, the membership hypotheses are discharged twice over: explicitly, with no K -theory, in Section 7; and via Proposition 8.3 here. Theorem B in its full strength—all ranks down to two, all finite fields, and the unit group itself—uses the second route.

9. FINITELY PRESENTED NONSOFTIC COVERS

Nonsficity of a finitely generated group is witnessed by a finite fragment of its multiplication table; imposing that fragment as a presentation produces finitely presented nonsfic covers. The covers are proved nonsfic directly, from the forbidden table itself; quotient closure of softicity, which is unknown, plays no role.

Definition 9.1. Let G be a group, $F \subseteq G$ finite with $1 \in F$, and $\varepsilon > 0$. An (F, ε) -model of G is a map $\varphi : F \cup F \cdot F \rightarrow \text{Sym}(Y)$, for some finite set Y , such that

$$(34) \quad d_H(\varphi(g)\varphi(h), \varphi(gh)) \leq \varepsilon \quad (g, h \in F),$$

$$(35) \quad d_H(\varphi(g), \varphi(h)) \geq 1 - \varepsilon \quad (g, h \in F, g \neq h).$$

Lemma 9.2 (Finite-table characterization). *A countable group G is softic if and only if it admits an (F, ε) -model for every finite $F \ni 1$ and every $\varepsilon > 0$.*

Proof. If σ_n is a softic approximation, then for fixed (F, ε) the restriction of σ_n to $F \cup F \cdot F$ is an (F, ε) -model for all large n : (34) is Definition 2.1(a), and (35) follows from (a) and (b) since $d_H(\sigma_n(g), \sigma_n(h)) \geq d_H(\sigma_n(gh^{-1})\sigma_n(h), \sigma_n(h)) - d_H(\sigma_n(gh^{-1})\sigma_n(h), \sigma_n(g)) = d_H(\sigma_n(gh^{-1}), \text{id}) - o(1)$.

Conversely, exhaust G by finite symmetric sets $F_1 \subseteq F_2 \subseteq \dots$ containing 1, and let φ_n be an $(F_n, 1/n)$ -model on Y_n ; replacing Y_n by many disjoint copies of itself we may assume $|Y_n| \rightarrow \infty$. Define $\sigma_n(g) = \varphi_n(g)$ for $g \in F_n$ and $\sigma_n(g) = \text{id}$ otherwise. Fix $g, h \in G$; for n large, $g, h, gh \in F_n$, and (34) gives Definition 2.1(a). For faithfulness, first note that $p_n = \varphi_n(1)$ satisfies $d_H(p_n^2, p_n) \leq 1/n$ by (34) with $g = h = 1$, and $d_H(p_n^2, p_n) = d_H(p_n, \text{id})$ by right invariance of d_H ; so $d_H(\varphi_n(1), \text{id}) \leq 1/n$. Then for $g \neq 1$ and n large,

$$d_H(\sigma_n(g), \text{id}) \geq d_H(\varphi_n(g), \varphi_n(1)) - d_H(\varphi_n(1), \text{id}) \geq 1 - \frac{2}{n},$$

by (35). Hence σ_n is a softic approximation. $\square$

Theorem 9.3 (Finite multiplication-table covers). *Let G be a finitely generated group that is not softic. Then there is an infinite finitely presented group H that is not softic, together with a surjection $H \twoheadrightarrow G$.*

Proof. Finite groups are softic, so G is infinite. By Lemma 9.2 there are a finite set $F \subseteq G$ with $1 \in F$ and an $\varepsilon_0 > 0$ such that G admits no (F, ε_0) -model; enlarging F , we may assume F is symmetric and generates G . Define the finitely presented group

$$(36) \quad H = H_F = \langle x_g \ (g \in F \cup F \cdot F) \mid x_1 = 1, \ x_g x_h = x_{gh} \ (g, h \in F) \rangle.$$

The assignments $x_g \mapsto g$ satisfy the relations in G , and F generates G , so there is a surjection

$$(37) \quad \pi : H \twoheadrightarrow G, \quad \pi(x_g) = g.$$

In particular H is infinite, and $\pi(x_g) = g \neq h = \pi(x_h)$ shows $x_g \neq x_h$ in H for distinct $g, h \in F \cup F \cdot F$.

TableCover
TableModel

TableCover
tableModel_of_isSoftic
exists_table_obstruction

TableCover
tableGroup_no_model
exists_finitelyPresented_
obstruction

Suppose H were sofic, with sofic approximation $\sigma_n : H \rightarrow \text{Sym}(Y_n)$. Define $\varphi_n : F \cup F \cdot F \rightarrow \text{Sym}(Y_n)$ by $\varphi_n(g) = \sigma_n(x_g)$. For $g, h \in F$, the relation $x_g x_h = x_{gh}$ holds in H , so

$$d_H(\varphi_n(g)\varphi_n(h), \varphi_n(gh)) = d_H(\sigma_n(x_g)\sigma_n(x_h), \sigma_n(x_g x_h)) \rightarrow 0.$$

For distinct $g, h \in F$, the element $x_g x_h^{-1} \neq 1$ of H gives, as in the proof of Lemma 9.2,

$$d_H(\varphi_n(g), \varphi_n(h)) \geq d_H(\sigma_n(x_g x_h^{-1}), \text{id}) - o(1) \rightarrow 1.$$

Hence for n large, φ_n is an (F, ε_0) -model of G —note that (34) and (35) refer only to the multiplication table of F inside G , which is faithfully copied by the generators of H . This contradicts the choice of (F, ε_0) . Therefore H is not sofic. $\square$

Theorem 9.4 (Kazhdan covers). *Let G be a nonsofic group with property (T). Then there is an infinite finitely presented property-(T) group H that is not sofic, together with a surjection $H \twoheadrightarrow G$.*

Proof. Property-(T) groups are finitely generated, and G is infinite as before. By Shalom’s theorem [19], G is a quotient of a finitely presented property-(T) group: fix a surjection $\theta : P \twoheadrightarrow G$ with P finitely presented and Kazhdan. Choose (F, ε_0) as in the proof of Theorem 9.3, and for each $g \in F \cup F \cdot F$ choose a lift $\tilde{g} \in P$ with $\theta(\tilde{g}) = g$, taking $\tilde{1} = 1$. Define

$$(38) \quad H = P / \langle\langle \tilde{g} \tilde{h} \tilde{gh}^{-1} : g, h \in F \rangle\rangle.$$

Then H is finitely presented (finitely many new relators over the finitely presented P) and has property (T) (a quotient of P). Each relator lies in $\ker \theta$, since $\theta(\tilde{g} \tilde{h} \tilde{gh}^{-1}) = gh(gh)^{-1} = 1$; hence θ descends to a surjection $H \twoheadrightarrow G$, and H is infinite. Writing $y_g \in H$ for the image of $\tilde{g}$, we have $y_1 = 1$, $y_g y_h = y_{gh}$ for $g, h \in F$, and $y_g \neq y_h$ for distinct $g, h \in F$ (their images in G differ). If H were sofic, then $\varphi_n(g) = \sigma_n(y_g)$ would be an (F, ε_0) -model of G for large n , exactly as in the proof of Theorem 9.3, a contradiction. $\square$

Proof of Theorem C. The first two assertions are Theorems 9.3 and 9.4. For the final assertion, apply Theorem 9.4 to the infinite finitely generated property-(T) nonsofic group $L^\times$ of Theorem B, over any fixed finite field; the resulting H surjects onto $L^\times$, and composing with the inverses of the isomorphisms Θ_C of Theorem B(ii) exhibits a quotient of H isomorphic to each group listed there. $\square$

Remark 9.5 (Effectivity). The covers are effective in the logical sense: non-soficity of G is equivalent, by Lemma 9.2, to the existence of a finite witness (F, ε_0) , and any witness yields H by (36) or (38). The proof establishes the existence of a witness without computing one, and no claim is made that $L^\times$ itself is finitely presented.

ShalomFinitePresentation
exists_presented_kazhdan_
cover
exists_finitelyPresented_
kazhdan_cover
KazhdanCover
exists_kazhdan_
finitelyPresented_cover_of_
not_isSofic
Shalom’s theorem is proved
internally, not cited.

10. CONCLUDING REMARKS

Remark 10.1 (The trust surface of Theorem A). Theorem A is independent of $\mathrm{GL}_r = \mathrm{EL}_r$ and of $K_1(L) = 0$: the memberships of u and z are explicit factorizations, the remaining memberships follow from the Whitehead identity and the perfectness of V , and the only analytic inequality proved in the paper is the co-area estimate of Lemma 2.7—the approximation-theoretic depth resides in the cited theorems of Kun and Kun–Thom. The argument concerns permutation models only; hyperlinearity of these groups, and the Connes embedding problem, remain open.

Remark 10.2 (Two compressors, one image). The pair $Q = \{u, v\}$ resolves a genuine tension: generation of G requires elements that move Γ , while compression requires elements that map Γ into itself. The present realization divides the two roles. The compressor u seats $K = u\Gamma u^{-1}$; the involution $z = vu^{-1}$ generates the missing last row and column of EL_n (Proposition 6.4); and the two compressors share the single image K , so the criterion sees one coherent compression while z carries the generation. The criterion itself is indifferent to the size of Q ; the algebra here simply supplies two natural compressors rather than one.

Remark 10.3 (The median is canonical up to scale). The normalization $f = s/(s + m)$ of Section 4 pins at $1/2$, and the value $1/2$ is not a trace or a spectral constant: it is the fixed point of the exchange of source and target sizes, and it is a vertex-weighted median by construction. Median pinning is exactly the statement that a permutation compressing one expander decomposition into another must asymptotically preserve the size profile—the conservative core of the argument, with exact conservation (Lemma 3.2) converting a one-sided drift estimate into a two-sided one at no cost.

Remark 10.4 (What property (T) is used for). Property (T) enters three times, in three different roles: (a) through Theorem 2.9 applied to Γ , producing the source decomposition that is compressed; (b) through Theorem 2.9 applied to G , producing the ambient decomposition carrying the co-area inequality—and by Remark 4.3 this use is replaceable by any componentwise ℓ^1 Poincaré decomposition; (c) through the Kazhdan factor hypothesis of Theorem 2.10. The local Theorem 4.1 quantifies over the first two uses and leaves only the third to the black box.

Remark 10.5 (Scope). The Leavitt-corner theorem (Theorem 6.7) applies to an arbitrary nontrivial unital ring carrying a binary Leavitt family, whenever the two adjacent elementary groups are Kazhdan and the two matrices u, z are elementary; for finitely generated rings and $m \geq 3$ the Kazhdan hypothesis is automatic (Corollary 6.8). Finite fields are where all hypotheses are verified simultaneously (Theorem 8.8); characteristic two is where they are verified explicitly (Theorem 7.3); and the construction can already be carried out in rank two (Theorem 8.1).

APPENDIX A. AN ELEMENTARY AND EFFECTIVE PROOF OF
 $K_1(L_k(1, 2)) = 0$

This appendix proves the vanishing of K_1 announced in Remark 8.5, in the Whitehead form in which Section 8 uses it, by elementary means: no localization sequence, no spectra, and no input beyond the two Leavitt relations and one classical fact about L recorded as Lemma A.3 below. The proof is an algorithm—a Gaussian elimination over the binary tree—and the complexity accounting of Remark 8.6 reads off from it. Every lemma of this appendix is formalized in the accompanying Lean development, and the margin notes record the correspondence; the appendix is written to be read and checked entirely on paper.

Throughout, k is a field and $L = L_k(1, 2)$. Recall from Section 5 the word notation s_w, s_w^* (we write $t_w = s_w^*$ for the word w , so $t_w s_{w'} = \delta_{w, w'}$ for $|w| = |w'|$), the cylinder idempotents $p_w = s_w t_w$, and complete leaf sets. A *complete code of depth* $\leq r$ (resp. $\geq r$) is a complete leaf set all of whose words have length $\leq r$ (resp. $\geq r$).

A.1. The Whitehead class group.

Definition A.1. Let

$$H = \{u \in L^\times \mid \text{diag}(u, 1) \in \text{EL}_2(L)\}.$$

Since $\text{diag}(uv, 1) = \text{diag}(u, 1) \text{diag}(v, 1)$, H is a subgroup of $L^\times$. The main theorem of this appendix is:

Theorem A.2. $H = L^\times$. *Equivalently, for every $u \in L^\times$ and every $r \geq 2$, the matrix $\text{diag}(u, 1, \dots, 1)$ is a product of elementary matrices in $\text{EL}_r(L)$.*

(The two forms are equivalent because $\text{EL}_2 \hookrightarrow \text{EL}_r$ in the first two coordinates.) The single external input is the following division property, a standard consequence of pure infiniteness and simplicity of L [1]; the accompanying formalization proves it from scratch, so a reader may take it either from the literature or from the machine-checked development.

Lemma A.3 (Strong division). *For every nonzero $x \in L$ there are $p, q \in L$ with $pxq = 1$.*

The first consequence is a Gaussian elimination for invertible matrices, which we prove at every rank; it is the GE property of Theorem 8.2(a) in elementary form.

Lemma A.4 (Elementary diagonalization). *Let A be a nontrivial unital ring in which every nonzero element x admits p, q with $pxq = 1$, and let $X \in \text{GL}_r(A)$, $r \geq 2$. Then there are $E, F \in \text{EL}_r(A)$ and $u \in A^\times$ with*

$$E X F = \text{diag}(u, 1, \dots, 1).$$

Proof. First let $r = 2$ and write $X = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. Assume first $a \neq 0$, and pick p, q with $paq = 1$. Three elementary operations plant a literal 1 in the

DiagonalClassGroup
stableUnits
ScalarReduction

RefineLoopDischarge
K1_trivial
narrowReduction_holds
scalarReduction_holds
stableUnits_eq_top_holds

LeavittSimplicity
exists_mul_mul_eq_one

formalized in part
MatrixDiagonalization
exists_elementary_mul_diag
binaryLeavitt_exists_
elementary_mul_diag
Rank two is formalized; the
higher-rank induction is
manuscript-only.

corner: the column operation $C_2 += C_1 \cdot \rho$ with $\rho = q(1 - pb)$ replaces b by $b' = a\rho + b$, which satisfies $pb' = paq(1 - pb) + pb = 1$; the row operation $R_2 += wR_1$ with $w = (1 - d')p$, where $d' = c\rho + d$, replaces d' by $wb' + d' = (1 - d')pb' + d' = 1$; and with a 1 at position $(2, 2)$, the operations $R_1 += (-b')R_2$ and $C_1 += C_2 \cdot (-c'')$ (where c'' is the current $(2, 1)$ entry) clear the off-diagonal entries. The result is $\text{diag}(v, 1)$ with $v \in A^\times$, since the product of invertible matrices is invertible and a triangular-free diagonal matrix is invertible exactly when its entries are. If $a = 0$, then some entry of X is nonzero (as X is invertible and $A \neq 0$), and one elementary row or column addition moves a nonzero element into position $(1, 1)$; this reduces to the previous case.

For $r > 2$, induct. Pick any nonzero entry of X and, by one row and one column addition, arrange $x_{11} \neq 0$. Run the four operations above in the coordinate pair $(1, r)$: they involve only rows and columns 1 and r , and they produce a literal 1 at position (r, r) . Using that 1, clear the rest of row r and column r by elementary operations. The matrix is now $\text{diag}(X', 1)$ with $X' \in \text{GL}_{r-1}(A)$, and the inductive hypothesis applies; for $r - 1 = 1$ the surviving corner is the unit u . $\square$

Lemma A.5. $\text{EL}_2(L)$ is normal in $\text{GL}_2(L)$, and H is normal in $L^\times$.

Proof. Conjugation by an elementary matrix preserves EL_2 ; conjugation by $\text{diag}(u, v)$ sends $x_{12}(a)$ to $x_{12}(uav^{-1})$ and $x_{21}(a)$ to $x_{21}(vau^{-1})$, hence also preserves EL_2 ; and by Lemma A.4 every element of $\text{GL}_2(L)$ is a product of elementary matrices and one matrix $\text{diag}(u, 1)$. For the second claim, if $u \in H$ and $g \in L^\times$, then $\text{diag}(gug^{-1}, 1) = \text{diag}(g, 1) \text{diag}(u, 1) \text{diag}(g, 1)^{-1} \in \text{EL}_2(L)$ by the first claim. $\square$

By Lemma A.5 the quotient $L^\times/H$ is a group; we write $[u]$ for the class of u , so that Theorem A.2 says $[u] = 1$ for every unit. Three families of moves generate equalities in $L^\times/H$.

Lemma A.6 (Unipotent moves). *Let $a, b \in L$ with $ba = 0$. Then $1 + ab \in H$. In particular, for prefix-incomparable words v, w and any $x \in L$, the cylinder unipotent $1 + s_v x t_w$ lies in H .*

Proof. From $ba = 0$, $(ab)^2 = a(ba)b = 0$, so $1 + ab$ is a unit with inverse $1 - ab$, and the four-term identity

$$\text{diag}(1 + ab, 1) = x_{12}(a) x_{21}(b) x_{12}(-a) x_{21}(-b)$$

holds by direct block computation: $x_{12}(a)x_{21}(b) = \begin{pmatrix} 1+ab & a \\ b & 1 \end{pmatrix}$, multiplying by $x_{12}(-a)$ clears the $(1, 2)$ entry using $-(1 + ab)a + a = -a(ba) = 0$, and $x_{21}(-b)$ clears the $(2, 1)$ entry. For the particular case take $a = s_v x$, $b = t_w$: then $ba = t_w s_v x = 0$ since v, w are incomparable. $\square$

Lemma A.7 (The flip). *Let $x, y \in L$ and suppose $1 + xy \in L^\times$. Then $1 + yx \in L^\times$, with inverse $1 - y(1 + xy)^{-1}x$, and*

$$[1 + xy] = [1 + yx] \quad \text{in } L^\times/H.$$

DiagonalClassGroup
conj_mem_elementaryGroup_of_
division
stableUnits_normal

StableUnitsGenerators
mem_stableUnits_of_val_
unipotent
IncomparableUnipotents
incomparableUnit_mem

WhiteheadFlip
flipUnit
diagPair_flipUnit_inv_mem
mul_flipUnit_inv_mem_
stableUnits

Proof. The stated inverse is checked directly: $(1 + xy)(1 - y(1 + xy)^{-1}x) = 1 + yx - y(1 + xy)(1 + xy)^{-1}x = 1$, and likewise on the other side. The matrix $\text{diag}(1 + xy, (1 + yx)^{-1})$ factors into four elementary matrices,

$$x_{12}(x) x_{21}(y) x_{12}(-(1 + xy)^{-1}x) x_{21}(-(1 + yx)y),$$

again by direct computation. Hence $\text{diag}((1 + xy)(1 + yx)^{-1}, 1) = \text{diag}(1 + xy, (1 + yx)^{-1}) \cdot \text{diag}((1 + yx)^{-1}, (1 + yx))^{-1}$. (a Whitehead diagonal, Lemma 5.5) lies in $\text{EL}_2(L)$, so $(1 + xy)(1 + yx)^{-1} \in H$. $\square$

Lemma A.8 (Corner insertion). *For a binary word w and $u \in L^\times$, the corner insertion $\kappa_w(u) = s_w u t_w + (1 - p_w)$ is a unit, and $[\kappa_w(u)] = [u]$. Consequently every central unit lies in H .*

*LeavittDiagonalClass
kappaUnit_mul_inv_mem_
stableUnits
central_mem_stableUnits
stableUnits_eq_top*

Proof. The matrix $X_w = \begin{pmatrix} s_w & 1-p_w \\ 0 & t_w \end{pmatrix}$ is invertible, with inverse $\begin{pmatrix} t_w & 0 \\ 1-p_w & s_w \end{pmatrix}$ (using $t_w s_w = 1$ and $(1 - p_w)s_w = 0$), and it intertwines:

$$X_w \text{diag}(u, 1) X_w^{-1} = \text{diag}(\kappa_w(u), 1).$$

Thus $\text{diag}(\kappa_w(u)u^{-1}, 1) = [X_w, \text{diag}(u, 1)]$, a commutator of invertible matrices, one of which is elementary-diagonalizable; by Lemmas A.4 and 5.5 such a commutator lies in $\text{EL}_2(L)$, so $\kappa_w(u)u^{-1} \in H$.

For a central unit c , centrality gives $s_0 c t_0 + s_1 c t_1 = c(s_0 t_0 + s_1 t_1) = c$; on the other hand a direct computation using $t_1 s_0 = 0$ shows

$$\kappa_0(c) \kappa_1(c) = s_0 c t_0 + s_1 c t_1.$$

Hence $[c] = [\kappa_0(c)][\kappa_1(c)] = [c]^2$ in $L^\times/H$, and $[c] = 1$. $\square$

A.2. Code moves. Complete leaf sets play the role of block structures. Two further families of units lie in H .

Lemma A.9 (Code scalars). *Let $\mathcal{D} = (w_1, \dots, w_\ell)$ be a complete leaf set and $G \in \text{GL}_\ell(k)$ an invertible scalar matrix. Then*

*CodeScalarMoves
codeScalar_unit_mem*

$$\Sigma_{\mathcal{D}}(G) = \sum_{i,j} s_{w_i} G_{ij} t_{w_j}$$

is a unit of L lying in H .

Proof. The assignment $G \mapsto \Sigma_{\mathcal{D}}(G)$ is unital and multiplicative by the leaf identities (Lemma 5.3), so $\Sigma_{\mathcal{D}}(G)$ is a unit with inverse $\Sigma_{\mathcal{D}}(G^{-1})$. Over the field k , G is a product of transvections and one diagonal matrix (Gaussian elimination). A transvection $I + \lambda E_{ij}$, $i \neq j$, transports to $1 + \lambda s_{w_i} t_{w_j}$, a cylinder unipotent in H by Lemma A.6 (leaves of a complete leaf set are pairwise incomparable). A diagonal matrix $\text{diag}(d_1, \dots, d_\ell)$ transports to $\prod_i \kappa_{w_i}(d_i)$ with central scalars $d_i \in k^\times$; each factor lies in H by Lemma A.8. $\square$

CodeChangeUnits
codeChange_mem_stableUnits
CodeChangeGlue
codeBijection_mem_
stableUnits

Lemma A.10 (Code changes). *Let $(\tau_1, \dots, \tau_\ell)$ and $(\sigma_1, \dots, \sigma_\ell)$ be complete leaf sets of the same size. Then the code change*

$$\Omega = \sum_{i=1}^{\ell} s_{\tau_i} t_{\sigma_i}$$

is a unit of L lying in H .

Proof. Ω is a unit with inverse $\sum_i s_{\sigma_i} t_{\tau_i}$, by the leaf identities. Note that the Ω are precisely the images U_g of Thompson’s group V under the embedding (23), together with their compositions with index bijections; the lemma is a generation theorem for the Higman–Thompson groupoid. Induct on ℓ . For $\ell = 1$ both sets are the empty word and $\Omega = 1$. For $\ell \geq 2$, the source set, being the leaf set of a finite full binary tree, contains a sibling pair $\sigma_a = w0$, $\sigma_b = w1$. Composing Ω with at most two *transpositions* aligns the pairing so that the targets of σ_a, σ_b are again a sibling pair $v0, v1$ in the same order; here a transposition is a code change that swaps two leaves x, y of one complete leaf set and fixes the rest, i.e. $T(x, y) = s_x t_y + s_y t_x + (1 - p_x - p_y)$, and $T(x, y) \in H$ because it is $\Sigma_{\mathcal{D}}(P)$ for the corresponding scalar permutation matrix P on any complete leaf set $\mathcal{D}$ refining $\{x, y\}$ —a code scalar (Lemma A.9). After alignment, the two sibling terms merge through the completeness relation, $s_{v0} t_{w0} + s_{v1} t_{w1} = s_v (s_0 t_0 + s_1 t_1) t_w = s_v t_w$, producing a code change between complete leaf sets of size $\ell - 1$; the inductive hypothesis and normality of H conclude. $\square$

PencilReshape
reshaped_pencil_mem_iff
exists_reshaped_pencil
pencilVal_window_mem

Corollary A.11 (Reshaping). *Let $(R_i), (C_j), (P_i), (Q_j)$ be complete leaf sets, with $|R| = |P|$ and $|C| = |Q|$, and let $E_{ij} \in L$. If*

$$u = \sum_{i,j} s_{R_i} E_{ij} t_{C_j} \quad \text{and} \quad v = \sum_{i,j} s_{P_i} E_{ij} t_{Q_j}$$

are both units, then $v = \Omega_1 u \Omega_2$ for code changes $\Omega_1 = \sum_i s_{P_i} t_{R_i}$ and $\Omega_2 = \sum_j s_{C_j} t_{Q_j}$, and hence $u \in H \iff v \in H$.

Proof. $\Omega_1 u \Omega_2 = \sum s_{P_i} (t_{R_i} s_{R_a}) E_{ab} (t_{C_b} s_{C_j}) t_{Q_j} = v$ by the leaf identities; apply Lemma A.10 and normality. $\square$

A.3. The degree filtration and width reduction. The defining relations of L are homogeneous for $\deg s_i = +1$, $\deg t_i = -1$, so the free-algebra grading descends: $L = \bigoplus_{d \in \mathbb{Z}} L_d$ is $\mathbb{Z}$ -graded, with L_d spanned by the monomials $s_\alpha t_\beta$, $|\alpha| - |\beta| = d$. Write

$$W[a, b] = \text{span}_k \{s_\alpha t_\beta \mid a \leq |\alpha| - |\beta| \leq b\},$$

and, for $x \in L$, write $x = \sum_d x^{(d)}$ for its graded components. Elements of $W[0, 0]$ are called *balanced*; by the leaf identities, the balanced elements supported at depth $\leq n$ are exactly the matrices $\sum_{\gamma, \delta} G_{\gamma\delta} s_\gamma t_\delta$ over the full code $\{0, 1\}^n$ with scalar G , i.e. the image of $M_{2^n}(k)$ under Σ -transport—so

Lemma A.9 says: *balanced units are exactly the transported invertible scalar matrices, and they lie in H^1 .*

Lemma A.12 (Width reduction). *Every $u \in L^\times$ satisfies $[u] = [u']$ for some unit u' with value in $W[-1, 1]$ (a narrow unit).*

LeavittWindowReduction
exists_window_reduction
LeavittGradingSpans
exists_corner_move
BinaryLeavittWindow
exists_narrow_representative

Proof. Every element of L lies in some window $W[-M, N]$. We show: if $N \geq 2$, then $[u] = [u']$ for a unit u' with value in $W[-M, \max(N - 1, 1)]$; the mirror statement for $M \geq 2$ follows by the same argument applied to the anti-automorphism θ of Lemma A.13 below, and iterating both lands in $W[-1, 1]$.

The engine is the *corner move*: for any $v, w \in L$,

$$\begin{bmatrix} u \end{bmatrix} = \begin{bmatrix} s_0(u + vw)t_0 + s_0vt_1 + s_1wt_0 + s_1t_1 \end{bmatrix},$$

obtained by transporting the 2×2 identity $\begin{pmatrix} 1 & v \\ 0 & 1 \end{pmatrix} \begin{pmatrix} u & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ w & 1 \end{pmatrix} = \begin{pmatrix} u+vw & v \\ w & 1 \end{pmatrix}$ along Θ_{C_2} of Proposition 5.4: the images of the two unipotent factors are the cylinder-type unipotents $1 + s_0vt_1$ and $1 + s_1wt_0$, which lie in H by Lemma A.6, and the image of $\text{diag}(u, 1)$ is $\kappa_0(u)$, congruent to u by Lemma A.8.

Let x be the value of u , with top component $x^{(N)}$, $N \geq 2$. Since $y t_0 s_0 = y$ for every y , the top component factors as $x^{(N)} = b s_0$ with $b = x^{(N)} t_0 \in W[N - 1, N - 1]$. Apply the corner move with $v = -b$, $w = s_0$: the new value is

$$s_0(x - x^{(N)})t_0 - s_0bt_1 + s_1s_0t_0 + s_1t_1.$$

The first summand lies in $W[-M, N - 1]$ (conjugation by $s_0(\cdot)t_0$ preserves degree), the second has degree $N - 1$, and the last two have degrees 1 and 0. Hence the new unit has value in $W[-M, \max(N - 1, 1)]$, as claimed. $\square$

A.4. Windows die. The next stage shows that units confined to a *one-sided* window already lie in H . This is where the analytic heart of the argument sits.

Lemma A.13 (Transpose). *The assignment $s_i \mapsto t_i$, $t_i \mapsto s_i$ extends to an anti-automorphism θ of L with $\theta^2 = \text{id}$; it maps $W[a, b]$ onto $W[-b, -a]$, and $u \in H \iff \theta(u) \in H$.*

ThetaStable
thetaMatUnit_mem_
elementaryGroup
thetaMatUnit_diagUnit

Proof. The relations are symmetric under the assignment composed with order reversal, so θ exists; it negates degree. Applying θ entrywise to the transpose of matrices sends $x_{ij}(a)$ to $x_{ji}(\theta(a))$ and $\text{diag}(u, 1)$ to $\text{diag}(\theta(u), 1)$, so it preserves EL_2 and hence membership in H . $\square$

Lemma A.14 (Pure tails are nilpotent). *Let $1 + \eta \in L^\times$ with $\eta \in W[1, 1]$. Then η is nilpotent.*

PureTailNilpotency
pure_tail_nilpotent

¹A balanced element that is a unit of L has invertible scalar matrix at its own depth—a singular matrix over a field is an annihilator of a nonzero matrix, hence a zero divisor—and therefore has balanced inverse.

Proof. Let $y = (1 + \eta)^{-1}$, with graded components $y^{(d)}$ supported in a finite window. The equation $(1 + \eta)y = 1$ reads degree-wise: $y^{(d)} + \eta y^{(d-1)} = \delta_{d,0}$. From the bottom of the window upward, $y^{(d)} = 0$ for d below the window; then $y^{(d)} = -\eta y^{(d-1)}$ for $d < 0$ forces all negative components to vanish, $y^{(0)} = 1$, and $y^{(d)} = (-\eta)^d$ for $d > 0$. Since the window is finite, $(-\eta)^d = 0$ for large d . $\square$

Lemma A.15 (Nilpotent tails die). *Let $z \in L$ be balanced with $s_1 z$ nilpotent, and let $u \in L^\times$ have value $1 + s_1 z$. Then $u \in H$.*

Proof. Induct on the nilpotency index D of $s_1 z$; for $D = 1$, $z = t_1(s_1 z) = 0$ and $u = 1$. For $D \geq 2$, put $r = t_1(s_1 z)^{D-1} \in W[D-2, D-2](\cdot)$; factoring $r = R\sigma$ with $R = rt_0^{D-2}$ balanced (as in Lemma A.12) and taking a balanced pseudo-inverse Ξ of R —balanced elements are scalar matrices along a full code, and every scalar matrix R has a matrix Ξ with $R\Xi R = R$ —one obtains an idempotent-like element e with $e(s_1 z) = 0$ and $(s_1 z)^{D-1}e = (s_1 z)^{D-1}$. The mover $1 - s_1(ze)$ has square-zero tail, hence lies in H by Lemma A.6, and

$$(1 + s_1 z)(1 - s_1(ze)) = 1 + s_1(z - ze)$$

because $ze s_1(ze)$ -cross terms vanish by $e(s_1 z) = 0$. The new tail satisfies $(s_1(z - ze))^{D-1} = (s_1 z)^{D-1}(1 - e) = 0$, so its index is at most $D - 1$, and the inductive hypothesis applies. $\square$

Lemma A.16 (The keystone). *Let $u = c + \zeta \in L^\times$ with c balanced and $\zeta \in W[1, 1]$. Then $c \in L^\times$.*

Proof. Suppose not. Balanced elements at depth n are scalar matrices $M_{2^n}(k)$ along the full code; by the rank normal form over the field k there are balanced units g, h with $gch = e := \sum_{\gamma \in S} p_\gamma$, a cylinder idempotent over a subset S of the depth- n code, and c is invertible exactly when S is everything. So assume S proper, and let $f = 1 - e = \sum_{\gamma \in T} p_\gamma$ with $T \neq \emptyset$. Replace u by $w = guh = e + \zeta'$ (with $\zeta' = g\zeta h \in W[1, 1]$); invertibility of c is not affected. Let $y = w^{-1}$, with graded components $y^{(d)}$, $d \in [\text{lo}, \text{hi}]$, and let $M = -\text{lo}$.

The equation $wy = 1$ reads degree-wise $e y^{(d)} + \zeta' y^{(d-1)} = \delta_{d,0}$. Multiplying the $d = 0$ equation by f on both sides ($fe = 0$, $f^2 = f$) gives $f = f\zeta' y^{(-1)} f$, and substituting the relation $y^{(d)} = f y^{(d)} - \zeta' y^{(d-1)}$ (valid for $d \leq -1$, from $e y^{(d)} = -\zeta' y^{(d-1)}$) repeatedly yields, after M steps (the components below the window vanish),

$$(39) \quad f = \sum_{j=0}^{M-1} \underbrace{f \zeta' (-\zeta')^j}_{\deg=j+1} \cdot \underbrace{f y^{(-1-j)} f}_{\deg=-1-j}.$$

Now count ranks at a deep uniform interface. For ℓ large, every factor in (39) is represented by a scalar matrix between the full codes of appropriate depths: an element of $W[d, d]$ supported at bounded depth is, for all large q ,

NilpotentTailKill
nilpotent_tail_mem_
stableUnits
RankNormalForm
exists_rank_normal_form

ZeroKOne
balanced_component_isUnit

a $2^{q+d} \times 2^q$ scalar matrix (pad with the completeness relation). Representing the j -th summand at interface depths $\ell \rightarrow \ell - 1 - j \rightarrow \ell$, the middle factor passes through the matrix of f at depth $\ell - 1 - j$, whose rank is exactly $|T| 2^{\ell-1-j-n}$. Hence the matrix of the right-hand side of (39) at depth ℓ has rank at most $\sum_{j < M} |T| 2^{\ell-1-j-n} < |T| 2^{\ell-n}$ —a geometric series short of its limit—while the matrix of f at depth ℓ has rank exactly $|T| 2^{\ell-n}$. This contradiction proves the lemma. $\square$

Proposition A.17 (Windows die). *A unit of L whose value lies in $W[0, N]$ for some N , or in $W[-N, 0]$, lies in H .*

WindowNonnegReduction
window_nonneg_mem_
stableUnits
WindowNonposReduction
window_nonpos_mem_
stableUnits
WidthTwoReduction
window_zero_one_mem_
stableUnits

Proof. By the transpose (Lemma A.13) it suffices to treat $W[0, N]$; induct on N .

Base, $N \leq 1$. Let $u = c + \zeta$, c balanced, $\zeta \in W[1, 1]$. By the keystone (Lemma A.16) c is a balanced unit, so $c \in H$ with balanced inverse, and $u = c(1 + \eta)$ with $\eta = c^{-1}\zeta \in W[1, 1]$. By Lemma A.14 η is nilpotent. The corner insertion $\kappa_1(1 + \eta) = 1 + s_1(\eta t_1)$ is congruent to $1 + \eta$ (Lemma A.8), has balanced tail coefficient $z = \eta t_1$, and $s_1 z = s_1 \eta t_1$ inherits nilpotency from η (indeed $(s_1 \eta t_1)^j = s_1 \eta^j t_1$); Lemma A.15 finishes.

Step, $N \geq 2$. Write the value as $1 + (a + \eta)$ with $a \in W[0, N - 1]$ —subtracting 1 costs nothing since 1 is balanced—and η the top component, of degree N . Split $\eta = s_0 q_0 + s_1 q_1$ with $q_z = t_z \eta$ of degree $N - 1$. Perform the stable Whitehead block move along the four depth-two cylinders: with

$$X_z = s_{00} s_z t_{w_z}, \quad Y_z = s_{w_z} q_z t_{00}, \quad (w_0, w_1) = (01, 10),$$

set $m_2 = (1 - X_0)(1 - X_1)$ and $m_1 = (1 + Y_0)(1 + Y_1)$, products of cylinder unipotents in H (Lemma A.6), and let $\kappa = \kappa_{00}(u) = 1 + s_{00}(a + \eta)t_{00}$, congruent to u . In the product $m_2 \kappa m_1$ all cross terms vanish by incomparability of the three cylinders 00, 01, 10, except

$$X_0 Y_0 + X_1 Y_1 = s_{00} (s_0 q_0 + s_1 q_1) t_{00} = s_{00} \eta t_{00},$$

which cancels the top component inside κ . The value of $m_2 \kappa m_1$ is therefore

$$1 + (s_{00} a t_{00} - X_0 - X_1 + Y_0 + Y_1),$$

whose components have degrees in $[0, N - 1]$: the a -part keeps its degrees, $\deg X_z = 1$, and $\deg Y_z = N - 1$. By induction $m_2 \kappa m_1 \in H$, hence $u \in H$. $\square$

A.5. The pencil and the elimination loop.

Lemma A.18 (Pencil form). *Let $x \in W[-1, 1]$. Then there are m and scalar matrices A_0, A_1, C, B_0, B_1 , indexed by the full code $\mathcal{F} = \{0, 1\}^{m+1}$, with*

PencilForm
exists_pencil_form
RefineLoopDischarge
pencilEntry_mem_window
codePair_expansion

$$x = \sum_{i,j \in \mathcal{F}} s_i E_{ij} t_j, \quad E_{ij} = A_{0,ij} t_0 + A_{1,ij} t_1 + C_{ij} \cdot 1 + B_{0,ij} s_0 + B_{1,ij} s_1.$$

Proof. Choose m with all monomials of x of depth $\leq m$, and expand $x = \sum_{i,j} s_i (t_i x s_j) t_j$ along the full code of depth $m + 1$ (completeness relation on both sides). For a monomial $s_\alpha t_\beta$ of x and code words i, j of length

$m + 1$: $t_i s_\alpha$ vanishes unless α is a prefix of i , in which case it is $t_{i'}$ with $|i'| = m + 1 - |\alpha| \geq 1$, and symmetrically $t_\beta s_j = s_{j'}$ with $|j'| = m + 1 - |\beta| \geq 1$; the product $t_{i'} s_{j'}$ collapses, by the prefix calculus, to 0, a scalar, or a single letter s_z or t_z , according to the degree $|\alpha| - |\beta| \in \{-1, 0, 1\}$. Hence each corner entry $t_i x s_j$ lies in the five-dimensional scalar pencil space displayed. $\square$

We call a unit in this form a *pencil unit* over the code pair (rows, columns); the data is the quintuple of scalar matrices, and the *stack* is the vertical block matrix $\begin{pmatrix} B_0 \\ B_1 \end{pmatrix}$.

Lemma A.19 (Strict negativity). *Let u be a pencil unit over complete leaf sets (R_i) , (C_j) , and suppose the stack admits a scalar left inverse: matrices G_0, G_1 with $G_0 B_0 + G_1 B_1 = I$. Then there is N such that every corner entry of the inverse,*

$$y_{ji} = t_{C_j} u^{-1} s_{R_i},$$

lies in $W[-N, -1]$.

Proof. All y_{ji} lie in a common finite window; let d^* be the largest degree in which any y_{ji} has a nonzero component, and suppose $d^* \geq 0$. From $u u^{-1} = 1$, sandwiching and inserting the completeness relation for (C_j) ,

$$\sum_j E_{ij} y_{ji'} = \delta_{ii'} \quad (\text{all } i, i').$$

Take the component of degree $d^* + 1 \geq 1$ on both sides. On the left only the degree-+1 part of E can contribute against $y^{(d^*)}$ —components of y above d^* vanish—so

$$\sum_j (B_{0,ij} s_0 + B_{1,ij} s_1) y_{ji'}^{(d^*)} = 0.$$

Multiplying on the left by t_0 and by t_1 separates the two shifts: $\sum_j B_{0,ij} y_{ji'}^{(d^*)} = 0$ and $\sum_j B_{1,ij} y_{ji'}^{(d^*)} = 0$; that is, the stack annihilates the matrix $Y = (y_{ji'}^{(d^*)})$ on the left. Applying the scalar left inverse, $Y = (G_0 B_0 + G_1 B_1) Y = 0$, contradicting the choice of d^* . Hence all components have degree ≤ -1 . $\square$

Lemma A.20 (Refinement step). *Let u be a pencil unit over (R, C) whose column j_0 is stack-free: $B_{0,ij_0} = B_{1,ij_0} = 0$ for all i . Then u is also a pencil unit over the code pair (R, C') , where C' replaces the word C_{j_0} by its two children $C_{j_0} 0, C_{j_0} 1$, with data given columnwise by*

$$(A_0, A_1, C, B_0, B_1)_{j_0 z} = (0, 0, A_{z, j_0}, [z=0] C_{j_0}, [z=1] C_{j_0}),$$

all other columns unchanged.

Proof. Insert $1 = s_0 t_0 + s_1 t_1$ before $t_{C_{j_0}}$: each term $s_{R_i} E_{ij_0} t_{C_{j_0}}$ becomes $\sum_z s_{R_i} (E_{ij_0} s_z) t_{C_{j_0} z}$, and for a stack-free entry the split identity

$$(a_0 t_0 + a_1 t_1 + c) s_z = a_z + c s_z$$

StrictNegativePencil
entry_window_negative_of_B_
full

RefineStep
refine_column
pencilEntry_mul_s
GLVectorNormalization
exists_isUnit_matrix_col

shows $E_{ij_0}s_z$ is again a pencil entry, with the A -data falling to the constant slot and the constant to the B -slot—the index shift of the Wiener–Hopf analogy. $\square$

Lemma A.21 (Code supply). *For $1 \leq \kappa \leq 2^r$ there is a complete leaf set of size κ and depth $\leq r$; for $\kappa \geq 2^M$ there is one of size κ and depth $\geq M$.*

CodeShapeSupply
exists_shallow_code
exists_deep_code

Proof. Kraft counting. Shallow: start from the full code of depth $\lceil \log_2 \kappa \rceil \leq r$ and merge sibling pairs until κ leaves remain (each merge reduces the count by one and cannot increase depth). Deep: start from the full code of depth M , of size $2^M \leq \kappa$, and split leaves (each split increases the count by one and produces words of length $\geq M$). $\square$

Theorem A.22 (The elimination loop). *Every pencil unit over a row code of 2^r words lies in H .*

RefineLoopDischarge
pencil_free_exit
pencil_full_exit
pencil_unit_mem_pow

Proof. Let u be a pencil unit over (R, C) , $|R| = 2^r$, $|C| = \kappa$. We induct on $2^{r+1} - \kappa$ (when this is ≤ 0 , the first case below applies).

Free exit ($\kappa \geq 2^{r+1}$). By Lemma A.21 choose a shallow complete code P of size 2^r and depth $\leq r$, and a deep complete code Q of size κ and depth $\geq r + 1$. Reshape u to the pair (P, Q) (Corollary A.11). Each pencil entry lies in $W[-1, 1]$, so each monomial of the reshaped value has degree at most $|P_i| + 1 - |Q_j| \leq r + 1 - (r + 1) = 0$: the reshaped unit lies in $W[-N', 0]$ for some N' , hence in H by Proposition A.17, hence $u \in H$.

Full stack. If the stack has a scalar left inverse, Lemma A.19 puts every corner entry y_{ji} of u^{-1} in $W[-N, -1]$. Expand $u^{-1} = \sum_{j,i} s_{C_j} y_{ji} t_{R_i}$ and reshape to a shallow column code of depth $\leq r + 1$ (possible: $\kappa \leq 2^{r+1}$ in this branch, or the free exit applies) and a deep row code of depth $\geq r$. Monomial degrees are at most $(r + 1) + (-1) - r = 0$, so $u^{-1} \in H$ by Proposition A.17, and $u \in H$.

Refine. Otherwise the stack, having no scalar left inverse, has a nonzero kernel vector v (over the field k , a matrix with trivial kernel has a left inverse). Choose an invertible scalar matrix G whose j_0 -th column is v (extend v to a basis). The code scalar $\Sigma_C(G)$ lies in H (Lemma A.9), and $u \cdot \Sigma_C(G)$ is a pencil unit over (R, C) with data multiplied columnwise by G ; its column j_0 is stack-free because v kills both B_0 and B_1 . Apply the refinement step (Lemma A.20): a pencil unit over (R, C') with $\kappa + 1$ columns. The counter $2^{r+1} - \kappa$ has dropped; the inductive hypothesis gives $u \cdot \Sigma_C(G) \in H$, hence $u \in H$. $\square$

Proof of Theorem A.2. Let $u \in L^\times$. By width reduction (Lemma A.12), $[u] = [u']$ with u' narrow; by the pencil form (Lemma A.18), u' is a pencil unit over the full code pair of depth $m + 1$, a row code of 2^{m+1} words; by the elimination loop (Theorem A.22), $u' \in H$. Hence $u \in H$. $\square$

A.6. Consequences and effectivity.

Corollary A.23. *For every field k and $L = L_k(1, 2)$: $K_1(L) = 0$; and $\text{GL}_r(L) = \text{EL}_r(L)$ for every $r \geq 2$.*

RefineLoopDischarge
glTwo_eq_elementary_holds
glFour_eq_elementary_holds

Proof. For $X \in \mathrm{GL}_r(L)$, Lemma A.4 writes $EXF = \mathrm{diag}(u, 1, \dots, 1)$, and Theorem A.2 places the diagonal factor in $\mathrm{EL}_r(L)$; hence $X \in \mathrm{EL}_r(L)$. The same normal form computes K_1 : every class of $\mathrm{GL}_r(L)/\mathrm{EL}_r(L)$ -stabilized is represented by $\mathrm{diag}(u, 1, \dots, 1)$, which is elementary. $\square$

The proof of Theorem A.2 is an algorithm, and every step is effective Gaussian elimination over k : the width reduction processes one graded component per corner move; the pencil form is a change of basis at depth $m + 1$; each loop pass computes a kernel vector or a left inverse of the stack; and both exits are reshapes along explicitly constructed codes followed by the window inductions, each step of which is again explicit. For a unit supported on monomials of depth at most d , all codes involved have at most 2^{d+2} words and the loop executes at most 2^{d+2} times, giving the complexity bound of Remark 8.6: the elementary factorization of $\mathrm{diag}(u, 1, \dots, 1)$ is computed in time polynomial in 2^d and in the number of monomials of u .

APPENDIX B. THE LEAN FORMALIZATION

Nothing in this appendix is used anywhere in the paper. The mathematics of Sections 1–10 is complete and self-contained as it stands, and a reader with no interest in machine verification may stop at Remark 10.5 without having skipped an argument. What follows records a second, independent check on the same mathematics, and says exactly how far that check reaches.

What the development is. The accompanying development

https://github.com/SauersML/nonsofic_existence

is a Lean 4 library (toolchain v4.32.2, on mathlib4 at the same tag) of roughly 260 modules. It is not a transcription of this paper. It is an independent formal proof of the same headline theorem, and in several places it proves more than the paper claims: Kun’s theorem (Theorem 2.9) and the Kun–Thom obstruction (Theorem 2.10) are cited here but *proved internally* there, and Proposition 5.12 is proved there for V itself, with no appeal to Higman’s presentation. Where the two differ, the margin notes say so.

The public endpoints are

```
nonsofic_groups_exist, universalLeavittEL4_not_isSofic,
exists_finitelyPresented_nonsofic_group,
```

all in the module `MainResults`. Each is a closed theorem: it takes no hypothesis naming Kun, Kun–Thom, or Ershov–Jaikin–Zapirain, and no `Nonempty` premise standing in for a construction. The property- (T) , compression, non-LEF, and expander-decomposition inputs are all built inside the library.

How to read the margin notes. Every numbered result of the paper—every definition, lemma, proposition, corollary, and theorem—carries a note in the margin. Remarks carry one only when they have formal content of their own; the concluding commentary does not.

A note in its plain form is a module name in *italic*, hyperlinked to its source file, followed in typewriter by the declarations in that module that carry the content. That is the whole claim: those declarations formalize this statement. A line underneath, when there is one, records a way in which the Lean statement is *stronger* than the one printed here—that Kun’s theorem is proved rather than assumed, say.

Two phrases override the plain reading, and they appear only where they are true.

formalized in part:

Only part of the statement is formalized. The line below says which part is, and which part is not.

not formalized:

The statement is not formalized, and the line below says why. This occurs exactly once, at Lemma 2.8, which the Lean proof does not need because it replaces the residual-finiteness route by a direct finite obstruction.

The notes are decoration in the strict sense: setting `\leanlinksfalse` in the preamble removes every one of them, leaving the text of the paper untouched.

The trust surface of the formalization. A machine proof is worth exactly what its axioms and its statements are worth, so both are audited mechanically rather than asserted.

Axioms. The module `Audit` prints the axiom report of every public endpoint on an ordinary build, and `scripts/Audit.lean` independently walks the transitive axiom closure of the whole project namespace and fails on anything outside classical Lean. Every endpoint reduces to `propext`, `Classical.choice`, and `Quot.sound`—the three axioms of classical mathematics in Lean. There are no `sorrys`, and the library builds with `warningAsError=true`.

Statements. An axiom audit cannot tell you that the theorem proved is the theorem you wanted; only reading the statements can. The declarations named in the margins are therefore the ones to read. Two definitional points carry most of the weight, and both are discharged as theorems rather than left as caveats: `isKazhdanPair_iff_complex` shows the real-orthogonal form of property (T) used in the library agrees with the complex-unitary form, with the same control set and tolerance, and `IsKazhdanPair.liftUniverse` removes the universe restriction, so `HasKazhdanPropertyT` is the textbook property.

Reproduction. `lake exe cache get && lake build` checks the library and prints the axiom reports. The margin notes are themselves checked on every build of this document by `scripts/check_lean_refs.py`, which fails if any note names a module or a declaration that the Lean source does not contain; so the correspondence recorded in the margins cannot silently rot as either side changes.

How the library is laid out. The margin notes give the statement-by-statement correspondence. The coarse structure behind them is the following; each layer is proved before the one below it uses it.

Principal modules	Role
Sofic, LEF, Kazhdan, ElementaryGroup	Hamming models, local embeddings, real-orthogonal property (T), and EL_n : the vocabulary of Section 2.
KunFiniteMarkov–KunDecomposition	Turns a property- (T) sofic model into negligible-edit uniform expander components: Theorem 2.9, proved rather than cited.
Refinement, Pinning, Selection, MedianNormalization	The five finite lemmas of Section 3 and the median normalization of Section 4.4.
CompressionSetup, Criterion, MatchingSelection, CriterionAssembly	Matches core and ambient components, localizes the product action, and closes Theorem 4.1 and Corollary 4.2.
KunThomFiniteMarkov–KunThomTheorem	Extracts LEF from exact-product expansion: Theorem 2.10, proved rather than cited.
FreeRoot*, A_2^* , FreeElementaryPropertyT	Property (T) for the free characteristic-two EL_3 , by an explicit A_2 spectral-gap computation: the input of Theorem 5.7.
UniversalLeavitt, UniversalRankFour, RankFourCompressors	The Leavitt relations, the two explicit compressors, and the corner witness: Sections 5–7.
RefineLoopDischarge	The two-exit elimination of Remark 8.5, giving $K_1(L) = 0$ in Whitehead form unconditionally.
MainResults, TableCover, Audit	Closes nonsoficity, builds the finitely presented cover (Theorem 9.3), and audits statements and axioms.

What is not formalized. Beyond Lemma 2.8, which is absent because the Lean proof does not need it, the gaps are those marked *formalized in part* in the margins: the arbitrary-finitely-generated-ring form of Theorem 5.7 (the characteristic-two cases the endpoints use are proved); part (b) of Theorem 8.2, which remains a cited input; the $\text{GL}_r = \text{EL}_r$ collapse of Proposition 8.3, formalized at ranks two and four rather than all ranks; the essential-expander definition (Definition 2.6), which the library realizes through the matching-certificate repair rather than as a definition of its own; and the complexity bounds of Remark 8.6, whose underlying elimination is formalized.

Two of the paper’s four external inputs are therefore cited here but proved there. Kun’s theorem and the Kun–Thom obstruction are the ones flagged in the margins at Theorems 2.9 and 2.10; Shalom’s theorem is the third, proved as `exists_finitelyPresented_kazhdan_cover`, so the Kazhdan

half of Theorem C rests on no external citation in the formalization either. Only property (T) for elementary groups over arbitrary finitely generated rings (Theorem 5.7) remains genuinely imported, and the endpoints use only its characteristic-two cases, which are proved.

It is worth being precise about what the last of these does and does not settle, because it is the one place where formalization changes the standing of a result in the body of the paper. The vanishing $K_1(L) = 0$ enters Theorem B through the localization sequence of [2]. The elementary elimination of Remark 8.5 is formalized unconditionally, in Whitehead form, as `K1_trivial`; so for the purposes of Theorem B that citation is confirmation rather than dependency.

REFERENCES

- [1] G. Abrams and G. Aranda Pino, *Purely infinite simple Leavitt path algebras*, J. Pure Appl. Algebra **207** (2006), 553–563.
- [2] P. Ara, M. Brustenga, and G. Cortiñas, *K-theory of Leavitt path algebras*, Münster J. Math. **2** (2009), 5–34.
- [3] P. Ara, K. R. Goodearl, and E. Pardo, *K_0 of purely infinite simple regular rings*, K-Theory **26** (2002), 69–100.
- [4] G. Baumslag and D. Solitar, *Some two-generator one-relator non-Hopfian groups*, Bull. Amer. Math. Soc. **68** (1962), 199–201.
- [5] C. Bleak and M. Quick, *The infinite simple group V of Richard J. Thompson: presentations by permutations*, Groups Geom. Dyn. **11** (2017), 1401–1436.
- [6] J. W. Cannon, W. J. Floyd, and W. R. Parry, *Introductory notes on Richard Thompson’s groups*, Enseign. Math. (2) **42** (1996), 215–256.
- [7] G. Elek and E. Szabó, *On sofic groups*, J. Group Theory **9** (2006), 161–171.
- [8] Kivanç Ersoy, *On minimal non-sofic and ω -non-sofic groups*, arXiv:2604.19174 [math.GR], 2026.
- [9] M. Ershov and A. Jaikin-Zapirain, *Property (T) for noncommutative universal lattices*, Invent. Math. **179** (2010), 303–347.
- [10] M. Gromov, *Endomorphisms of symbolic algebraic varieties*, J. Eur. Math. Soc. (JEMS) **1** (1999), 109–197.
- [11] G. Higman, *Finitely Presented Infinite Simple Groups*, Notes on Pure Mathematics, vol. 8, Department of Pure Mathematics, Department of Mathematics, I.A.S., Australian National University, Canberra, 1974.
- [12] H. V. Khanh and V. H. Thanh, *Matrix generators for the unit groups of $L_K(1, d)$* , arXiv:2607.10351 [math.RA], 2026.
- [13] G. Kun, *On sofic approximations of Property (T) groups*, arXiv:1606.04471v5 [math.CO], 2016; revised 2019.
- [14] G. Kun and A. Thom, *Inapproximability of actions and Kazhdan’s property (T)*, arXiv:1901.03963v2 [math.GR], 2019.
- [15] W. G. Leavitt, *The module type of a ring*, Trans. Amer. Math. Soc. **103** (1962), 113–130.
- [16] V. Nekrashevych, *Cuntz–Pimsner algebras of group actions*, J. Operator Theory **52** (2004), 223–249.
- [17] E. Pardo, *The isomorphism problem for Higman–Thompson groups*, J. Algebra **344** (2011), 172–183.
- [18] V. G. Pestov, *Hyperlinear and sofic groups: a brief guide*, Bull. Symbolic Logic **14** (2008), 449–480.

- [19] Y. Shalom, *Rigidity of commensurators and irreducible lattices*, Invent. Math. **141** (2000), 1–54.
- [20] A. M. Vershik and E. I. Gordon, *Groups that are locally embeddable in the class of finite groups*, Algebra i Analiz **9** (1997), 71–97; English transl., St. Petersburg Math. J. **9** (1998), 49–67.
- [21] B. Weiss, *Sofic groups and dynamical systems*, Sankhyā Ser. A **62** (2000), 350–359.